# SOME PROBLEMS IN COMPLEX VARIABLES

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Abstract. The purpose of this survey is to collect problems in contemporary several complex variables and complex geometry. The problems are drawn from [1, 2, 3, 4], consolidated, and organized by subject and degree of resolution. Each section introduces shared notation; proof sketches summarize established results, and attributed remarks record perspectives from the literature.

## Introduction

Several complex variables grew from analytic continuation and the geometry of domains of holomorphy. Hartogs extension revealed a fundamental distinction from one variable: if $n \geq 2$, $\Omega \subset \mathbb{C}^n$ is a domain, and $K \Subset \Omega$ is compact with $\Omega \setminus K$ connected, every $f \in \mathcal{O}(\Omega \setminus K)$ extends to $\Omega$. The Levi problem connected holomorphic convexity with pseudoconvexity. Oka, Cartan, and Grauert placed these questions in the broader setting of Stein spaces, coherent sheaves, and complex manifolds [1, 2, 3].

The $L^2$ methods of Hörmander and Kohn made estimates for $\bar{\partial}$ central to existence, regularity, and extension. Kodaira's embedding theory connected analytic positivity with algebraic geometry. Pluripotential theory, invariant metrics, and holomorphic dynamics now link these foundations to hyperbolicity, algebraic cycles, and period maps [3, 4].

Our starting points are the problem lists of Buzzard–Fornæss–Gavosto–Krantz [1] and Fornæss–Sibony [2], Zhou's survey [3], and Siu's conference report [4]. We combine overlapping questions and retain classical problems whose solutions clarify the hypotheses or methods of unresolved ones.

We classify the problems into three categories: wide-open problems, partially resolved problems, and closed problems. By a closed problem, we mean one that has been completely resolved, either affirmatively or negatively. By a partially resolved problem, we mean one for which significant special cases or weaker forms are known, while the full statement remains open. By a wide-open problem, we mean one whose central question remains largely unresolved, although related results and proposed approaches may be available. The classification concerns the precise formulations stated here and reflects the cited literature.

**Reading guide.** Each subject is divided into closed problems with complete solutions, open problems with partial results, and wide-open problems, in that order. Problem $i.j$ is the $j$th problem of Section $i$, independently of its subsection. Proofs may be sketches; partial proofs specify the hypotheses of the known results, and remarks are attributed to their sources. Dates refer to cited publications unless otherwise stated, and recent preprint claims are identified explicitly.

Throughout, $\Delta = \{z \in \mathbb{C} : |z| < 1\}$, $\mathbb{B}^n = \{z \in \mathbb{C}^n : \|z\| < 1\}$, and $\mathbb{P}^n$ is complex projective space. The notation $A \Subset B$ means that $\overline{A}$ is compact and contained in $B$, and $\mathcal{O}(X)$ denotes the holomorphic functions on $X$. Smooth means $C^\infty$ unless otherwise specified. In algebraic

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parameter spaces, generic means outside a proper Zariski-closed subset; very general means outside a countable union of such subsets. Other notions of genericity are specified where used.

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## 1. Stein geometry and holomorphic embeddings

The Levi problem relates local pseudoconvexity to global holomorphic convexity. Its extensions to singular spaces, increasing unions, and embeddings remain central to Stein geometry. Here $\mathcal{O}(X)$ is the ring of holomorphic functions on $X$, and $\Omega^p_X$ is the sheaf of holomorphic $p$-forms. For compact $K \subset X$,

$$\widehat{K}_X = \{x \in X : |f(x)| \le \sup_K |f| \text{ for every } f \in \mathcal{O}(X)\}.$$

For a subset of $\mathbb{C}^N$, an unsubscripted $\widehat{K}$ denotes its polynomial hull. We write $\kappa_X$ for the infinitesimal Kobayashi pseudometric and $X \cong Y$ for biholomorphic equivalence. A domain $U \subset X$ is *locally Stein at the boundary* if every point of $\partial U$ has a neighborhood $V$ for which $U \cap V$ is Stein. An inclusion $U \subset V$ of Stein spaces is *Runge* if $\mathcal{O}(V)|_U$ is dense in $\mathcal{O}(U)$ in the compact-open topology. A *long* $\mathbb{C}^n$ is an increasing union of open subsets biholomorphic to $\mathbb{C}^n$.

### 1.1. **Closed problems with complete solutions.**

**Problem 1.1** (Levi problem)**.** Is every pseudoconvex domain $D \subset \mathbb{C}^n$ Stein? More generally, is a complex manifold $X$ Stein if and only if it admits a smooth strictly plurisubharmonic exhaustion $\rho : X \to \mathbb{R}$?

*Proof.* Oka (1953) proved the domain assertion [134], and Grauert (1958) proved the characterization by a smooth strictly plurisubharmonic exhaustion [105]. If $F : X \hookrightarrow \mathbb{C}^N$ is a proper holomorphic embedding of a Stein manifold, then $\rho = \|F\|^2$ is such an exhaustion. Conversely, Grauert's construction uses strongly pseudoconvex sublevel sets to establish holomorphic convexity and construct holomorphic functions separating points and giving local coordinates. Hence $X$ is Stein. □

**Problem 1.2** (Cousin problems)**.** Let $X$ be a Stein space. Is every additive Cousin datum globally solvable? Which Cartier divisors on $X$ are principal?

*Proof.* Cartan's Theorem B (1952), in the Stein-space formulation presented by Grauert and Remmert (1979), gives $H^q(X, \mathcal{O}_X) = 0$ for $q > 0$ [135, 106]. Thus every additive Cousin cocycle is a coboundary. The exponential sequence gives

$$\mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*) \cong H^2(X, \mathbb{Z}).$$

A Cartier divisor is principal exactly when its associated line bundle is holomorphically trivial; the displayed isomorphism makes this equivalent to topological triviality. □

**Problem 1.3.** Let $X = \bigcup_{j\geq 1} X_j$ be a complex surface, where $X_j \subset X_{j+1}$ and $X_j \cong \mathbb{B}^2$. Must $X$ be biholomorphic to $\mathbb{B}^2$, $\Delta \times \mathbb{C}$, or $\mathbb{C}^2$? Must every long $\mathbb{C}^2$ be biholomorphic to $\mathbb{C}^2$?

*Proof.* Wold (2010) constructed a non-Stein long $\mathbb{C}^2$, also exhaustible by balls [6]. Iterated non-Runge embeddings force the holomorphic hull of a compact set to escape every compact stage. Since $\mathbb{B}^2$, $\Delta \times \mathbb{C}$, and $\mathbb{C}^2$ are Stein, this gives negative answers to both questions. Boc Thaler and Forstnerič (2016) further constructed long $\mathbb{C}^2$'s without nonconstant holomorphic or plurisubharmonic functions [45]. □

**Problem 1.4.** Let $n \geq 2$ and let $X = \bigcup_{j\geq 1} X_j$ be a complex manifold, where the $X_j$ are open, $X_j \subset X_{j+1}$, and $X_j \cong \mathbb{B}^n$. If $\kappa_X \equiv 0$, must $X \cong \mathbb{C}^n$?

*Proof.* Fornæss (2004) constructed short $\mathbb{C}^n$ as non-autonomous basins exhausted by balls, with identically vanishing Kobayashi pseudometric and a bounded nonconstant plurisubharmonic function [42]. Such a function cannot exist on $\mathbb{C}^n$, since its restriction to every complex line is constant by the subharmonic Liouville theorem. Consequently these basins are counterexamples. Fornæss and Pal (2021 preprint) discuss further properties of the construction [43]. □

**Problem 1.5.** Does there exist a pseudoconvex domain $D \subset \mathbb{C}^2$ homotopy equivalent to $S^2$, written $D \simeq S^2$?

*Proof.* Gompf (2013) constructed domains with the required homotopy type, including examples with smooth compact boundary [7]. He constructs a suitable smoothly embedded handlebody and applies his isotopy criterion to make its interior Stein while preserving its homotopy type. □

**Problem 1.6.** Let $\pi : \mathbb{C}^2 \to \mathbb{C}$ be the first-coordinate projection and let $Y \subset \mathbb{C}^2$ be compact, with $\pi(Y) \subset \partial\Delta$. Must every open neighborhood $U \supset \widehat{Y}$ contain a Riemann surface $S$ such that $\pi|_S : S \to \Delta$ is a finite branched covering?

*Proof.* Alexander and Wermer posed the unrestricted formulation in the 1995 problem list [1, Problem 11.2.1]. It has the following elementary counterexample: take $Y = \{(1, 0)\}$, so $\widehat{Y} = Y$, and take a small ball $U$ about $(1, 0)$ whose first-coordinate projection misses 0. No surface contained in $U$ can project surjectively onto $\Delta$. □

## 1.2. **Open problems with partial results.**

**Problem 1.7** (Levi problem on singular spaces)**.** Let $X$ be a Stein space and $U \subset X$ open. If $U$ is locally Stein at every point of $\partial U$, must $U$ be Stein?

*Partial proof.* Docquier and Grauert (1960) proved the assertion for Stein manifolds by retracting a Stein neighborhood of the ambient manifold in Euclidean space. Andreotti and Narasimhan (1964) proved it for Stein spaces with isolated singularities by patching boundary-distance functions to construct a strictly plurisubharmonic exhaustion; Colţoiu (2009) surveys these results and methods [54]. Zhang (2021) proves the case $U = X \setminus A$ with $A$ a closed analytic subspace, assuming that this complement is locally Stein [18]. □

**Problem 1.8** (Union problem)**.** Let $X$ be a Stein space and $U_1 \subset U_2 \subset \cdots \subset X$ Stein open subsets. Must $\bigcup_{j\geq 1} U_j$ be Stein?

*Partial proof.* Behnke and Stein (1939) proved the increasing-union theorem for domains of holomorphy in $\mathbb{C}^n$. Together with the Stein-manifold Levi theorem of Docquier and Grauert (1960), it gives the nonsingular case described by Colţoiu (2009) [54]. Indeed, in a coordinate ball about a boundary point of $U = \bigcup_j U_j$, the increasing intersections with $U_j$ are pseudoconvex, and their union is pseudoconvex. Hence $U$ is locally Stein, and the Docquier–Grauert theorem makes $U$ Stein. □

**Problem 1.9.** Let $X$ be a Stein $n$-fold with $TX \cong X \times \mathbb{C}^n$ holomorphically. Does there exist a holomorphic map $F : X \to \mathbb{C}^n$ such that $\operatorname{rank} dF_x = n$ for every $x \in X$?

*Partial proof.* Gunning and Narasimhan (1967) proved the case $n = 1$ by constructing a noncritical holomorphic function on every open Riemann surface [136]. Forstnerič (2003) proved a submersion h-principle for maps from a Stein $n$-fold to $\mathbb{C}^q$ when $q < n$ [28]. A trivial holomorphic tangent bundle provides the formal epimorphism required by that theorem, so a parallelizable Stein $n$-fold has a holomorphic submersion to $\mathbb{C}^{n-1}$ for $n > 1$. □

**Problem 1.10.** Does every open Riemann surface admit a proper holomorphic embedding into $\mathbb{C}^2$?

*Partial proof.* Forstnerič and Wold (2009) proved proper embeddability for a bordered Riemann surface whose closure admits an injective immersion into $\mathbb{C}^2$ holomorphic in its interior [119]. They expose boundary points and apply automorphisms to push successive boundary pieces to infinity. Forstnerič and Wold (2013) proved proper embeddability for every circled domain in the Riemann sphere [120]. Alarcón and Forstnerič (2024) obtained further interpolation results and, in every compact Riemann surface, a Cantor set whose complement embeds properly in $\mathbb{C}^2$ [121]. □

**Problem 1.11.** Characterize compact complex manifolds $X$ for which every proper domain $D \subsetneq X$ that is locally Stein at the boundary is Stein. If $X$ is smooth projective, $D$ is locally Stein at the boundary, and $\partial D$ has a strongly pseudoconvex point, must $D$ be holomorphically convex?

*Partial proof.* Takeuchi (1964) proved that every proper pseudoconvex domain in $\mathbb{P}^n$ is Stein [137]. The Fubini–Study metric supplies strict plurisubharmonicity for a suitable boundary-distance exhaustion, and the Levi theorem then yields Steinness. Thus $X = \mathbb{P}^n$ satisfies the first assertion, and the domain in the second assertion is Stein, hence holomorphically convex, in this ambient case. Siu (1978) discusses this projective Levi theory [73]. □

**Problem 1.12.** Let $D \subsetneq X$ be a domain with connected smooth boundary in a compact complex manifold. Find geometric hypotheses under which every function holomorphic near $\partial D$, or every $C^1$ CR function on $\partial D$, extends holomorphically to $D$ or $X \setminus \overline{D}$.

*Partial proof.* Sarkis (2003), following his 2000 preprint, proved a one-sided Hartogs–Bochner theorem for a connected $C^2$ hypersurface $M \subset \mathbb{P}^n$, $n \geq 2$, separating projective space into two connected domains and having at most one open CR orbit [138]. There is one side to which every $C^1$ CR function on $M$ extends holomorphically. The proof combines propagation along CR orbits with holomorphic extension and complex Plateau methods. The CR-orbit hypothesis is part of the published theorem. □

**Problem 1.13.** Let $M \subset \mathbb{C}^n$ be a smooth compact totally real submanifold with $\dim_{\mathbb{R}} M = n$. Does there exist a compact bordered Riemann surface $\overline{R}$ and a nonconstant map $f \in C^\infty(\overline{R}, \mathbb{C}^n)$, holomorphic on $R$, such that $f(\partial R) \subset M$?

*Partial proof.* Gromov (1985) proved the existence of nonconstant attached holomorphic discs for compact Lagrangian submanifolds of standard symplectic $\mathbb{C}^n$ [72]. His method deforms a Cauchy–Riemann equation; compactness fails through disc bubbling, producing a disc with boundary on the Lagrangian submanifold. Boundary regularity gives smooth attachment when the submanifold is smooth. □

**Problem 1.14** (Oka–Grauert principle)**.** Develop quantitative and singular versions of the Oka–Grauert principle: specify hypotheses on a holomorphic family $\pi : Z \to X$ over a Stein space under which continuous sections can be deformed through sections to holomorphic sections with prescribed growth, boundary regularity, or singularity conditions.

*Partial proof.* Forstnerič (2010) proved the parametric Oka principle with approximation and jet interpolation for sections of stratified holomorphic fibre bundles over reduced Stein spaces whose fibres satisfy the convex approximation property [92]. Forstnerič (2026, final preprint accepted for publication) proved that, on a smoothly bounded strongly pseudoconvex Stein domain, a continuous section of a bundle of Hölder–Zygmund class $\Lambda^r$, $r > 0$, with Oka fibre is homotopic to a section of that class holomorphic in the interior [93].

In their 2026 preprint, An, Du, Guo, Wang, and Xie prove that every smooth projective rationally connected manifold is Oka. For a smooth hypersurface $D \subset \mathbb{P}^n$ of degree $d \geq 1$, they also prove

$$\mathbb{P}^n \setminus D \text{ is Oka} \quad \Longleftrightarrow \quad d \leq n+1$$

[146, Theorems 1.10–1.11]. Combined with the section theorems above, these results supply further admissible fibres; they do not remove the bundle hypotheses or give arbitrary prescribed growth. □

*Remark.* An, Du, Guo, Wang, and Xie explain that their local-to-global approximation argument builds on OpenAI's K3 manuscript [147], while their contributions include a criterion in arbitrary dimension and additional analytic refinements [146, §1, "Relation to the work of OpenAI"].

**Problem 1.15.** Let $X$ be a complex manifold satisfying

$$H^q(X, \Omega_X^p) = 0 \qquad (p \geq 0,\ q > 0).$$

Must $X$ be Stein?

*Partial proof.* Berg (1985) proved that a relatively compact open subset of a Stein space is a domain of holomorphy if $H^q(X, \mathcal{O}_X) = 0$ for all $q > 0$ [107]. In particular, when $X \Subset \mathbb{C}^n$, the displayed hypothesis with $p = 0$ makes $X$ a domain of holomorphy and hence Stein. □

## 1.3. **Wide-open problems.**

**Problem 1.16.** Let $D \Subset \mathbb{C}^n$ be a smoothly bounded domain with envelope of holomorphy $\pi : \widetilde{D} \to \mathbb{C}^n$. Is $\pi$ finitely sheeted? Is

$$\mathrm{Vol}(\widetilde{D}) := \int_{\widetilde{D}} \pi^*(dV) < \infty,$$

where $dV$ is Euclidean volume on $\mathbb{C}^n$?

**Problem 1.17.** Let $D \subset \mathbb{C}^2$ be strictly pseudoconvex and diffeomorphic to $\mathbb{R}^4$. Let $A$ be an embedded analytic disc and $M$ an embedded smooth disc such that

$$A \setminus \partial A \subset \mathbb{C}^2 \setminus D, \qquad M \setminus \partial M \subset D, \qquad \partial A = \partial M \subset \partial D.$$

If $M$ is totally real except at one elliptic complex point, must its envelope of holomorphy contain $A$?

**Problem 1.18.** Let $X \subset \mathbb{B}^N$ be an analytic subvariety with an isolated singularity at 0. Let $(\Omega_t)_{0\le t\le 1}$ be a continuously increasing family of Stein open subsets of $X$, with $0 \in \partial\Omega_{1/2}$ and $0 \in \Omega_t$ for $t > 1/2$. Is $\Omega_s$ Runge in $\Omega_t$ whenever $s < t$?

*Remark.* Buzzard, Fornæss, Gavosto, and Krantz (1995) record an equivalent polynomial-hull formulation: if $K \subset X \setminus \{0\}$ is compact and $0 \in \widehat{K}$, must every $L \subset X$ with $K \subset \operatorname{int}_X L$ satisfy $0 \in \operatorname{int}_X(\widehat{L} \cap X)$? Both hulls are taken in the ambient $\mathbb{C}^N$, and the interiors are relative to $X$ [1, Problems 4.2.1–4.2.2].

**Problem 1.19.** Let $\Omega \Subset \mathbb{C}^n$ be smoothly bounded and pseudoconvex. For compact $K \subset \overline{\Omega}$, define

$$\widehat{K}^{\mathrm{psh}} = \bigcap_{u\in \mathrm{PSH}(\Omega)\cap C(\overline{\Omega})} \{z \in \overline{\Omega} : u(z) \le \sup_K u\}.$$

If $\widehat{K}^{\mathrm{psh}} \cap \partial\Omega = K \cap \partial\Omega$ for every such $K$, must $\overline{\Omega}$ have a Stein neighborhood basis?

**Problem 1.20.** Does every compact reduced complex space $X$ admit a holomorphic embedding $X \hookrightarrow M$ into a complex manifold? Can $M$ always be chosen compact?

**Problem 1.21.** Does there exist a smoothly bounded strictly pseudoconvex domain in a complex surface in $\mathbb{P}^3$ that meets every compact complex curve in $\mathbb{P}^3$? If $V \subset \mathbb{P}^3$ is an algebraic hypersurface and $M \subset V$ is strictly pseudoconvex, must there exist a compact complex curve $S \subset \mathbb{P}^3$ with $S \cap V \subset V \setminus M$?

*Remark.* J. E. Fornæss and N. Sibony (2001) remark that B. Fabre constructed a real three-dimensional subset of $\mathbb{P}^3$ meeting every compact Riemann surface in $\mathbb{P}^3$; they refer to page 88 of Fabre's 2000 thesis [2, Question 3.11].

## 2. $L^2$ theory and CR geometry

The $\bar\partial$ equation connects holomorphic extension, boundary regularity, and positivity. Unless otherwise stated, $D \subset \mathbb{C}^n$ is a domain, data are $(0,1)$-forms closed in the distributional sense, and unweighted norms use Euclidean volume and the Euclidean metric. We write $N_q$ for the $L^2$ $\bar\partial$-Neumann operator on $(0,q)$-forms and $\Box_b$ for the tangential Kohn Laplacian. The spaces $H^s$ and $\Lambda^\alpha$ denote Sobolev and Hölder–Zygmund spaces; for $0 < \alpha < 1$, $\Lambda^\alpha$ is the usual Hölder space. Solution estimates have constants independent of the datum. On a smooth bounded pseudoconvex domain, $N_q$ is the inverse of $\Box_q = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$ on $(\ker \Box_q)^\perp$, with the $L^2$ adjoint and its natural operator domain. Finite type means D'Angelo finite type when no other notion is specified. For a plurisubharmonic function $\varphi$, its multiplier ideal is

$$\mathcal{I}(\varphi) = \{f \in \mathcal{O} : |f|^2 e^{-2\varphi} \in L^1_{\mathrm{loc}}\}.$$

### 2.1. Closed problems with complete solutions.

**Problem 2.1** (Ohsawa–Takegoshi extension problem)**.** Let $D \subset \mathbb{C}^{n-1} \times \Delta$ be pseudoconvex, let $H = D \cap \{z_n = 0\} \neq \varnothing$, and let $\varphi \in \mathrm{PSH}(D)$ satisfy $\varphi|_{H_0} \not\equiv -\infty$ on each component $H_0$ of $H$. Does every $f \in \mathcal{O}(H)$ with $\int_H |f|^2 e^{-\varphi} < \infty$ admit $F \in \mathcal{O}(D)$ such that $F|_H = f$ and

$$\int_D |F|^2 e^{-\varphi}\, dV_{2n} \le \pi \int_H |f|^2 e^{-\varphi}\, dV_{2n-2}?$$

*Proof.* Guan and Zhou (2015) prove the estimate with the optimal constant $\pi$ [110]. They choose auxiliary weights and twisting factors through an ordinary differential equation and apply the twisted Bochner–Kodaira identity. Zhou and Zhu (2018) extend the optimal extension method to weakly pseudoconvex Kähler manifolds under curvature hypotheses [114]. □

**Problem 2.2** (Strong openness conjecture)**.** Does every plurisubharmonic germ $\varphi$ satisfy

$$\mathcal{I}(\varphi) = \bigcup_{\varepsilon>0} \mathcal{I}((1+\varepsilon)\varphi)?$$

*Proof.* Guan and Zhou (2015) prove the equality [112]. Their proof combines induction and curve selection with $L^2$ extension from moving slices and a one-variable lower norm estimate, yielding a strict margin of integrability. □

**Problem 2.3.** Let $u, v$ be plurisubharmonic germs at a point $x$ of a complex manifold. Is $\mathcal{I}(tu) = \mathcal{I}(tv)$ for every $t > 0$ equivalent to $\nu(u\circ\pi, y) = \nu(v\circ\pi, y)$ for every proper modification $\pi : \widetilde{X} \to X$ with $\widetilde{X}$ smooth and every $y \in \pi^{-1}(x)$, where $\nu$ denotes the Lelong number?

*Proof.* Boucksom, Favre and Jonsson (2008) establish the valuative characterization for regularized multiplier ideals [113]. Guan and Zhou (2015), Corollary 3.3, use strong openness to identify these with the usual multiplier ideals, giving the stated equivalence [112]. □

**Problem 2.4.** Let $L$ be a big line bundle on a smooth projective variety, and let $h_{\min}$ be a semipositively curved singular metric with minimal singularities. Is

$$\mathcal{I}(\|mL\|) = \mathcal{I}(h_{\min}^m) \qquad (m \geq 1),$$

where the left side is the asymptotic multiplier ideal and $\mathcal{I}(h_{\min}^m) = \mathcal{I}(m\varphi_{\min})$ in a local trivialization $h_{\min} = e^{-2\varphi_{\min}}$?

*Proof.* Guan and Zhou (2015) derive this equality from strong openness and the comparison with asymptotic multiplier ideals [112, §3.4.1]. Increasing the singularity coefficient slightly supplies the reverse inclusion; the other inclusion is the standard comparison with a metric of minimal singularities. □

**Problem 2.5.** Let $\pi : X \to Y$ be a proper surjective holomorphic map from a Kähler manifold to a complex manifold, and let $(L, h)$ be a line bundle with a semipositively curved singular metric. Fix $m \geq 1$. On smooth fibres, form the relative $m$-Bergman kernel using sections of $mK_{X_y} + L|_{X_y}$ with fibre $L^{2/m}$ integral at most one. Assume this kernel is not identically zero, and extend its inverse as a semipositively curved metric on $mK_{X/Y} + L$. On a smooth exceptional fibre, does this extended metric equal the inverse extremal kernel obtained using only sections that extend to a neighbourhood of the fibre? Here exceptional means that not every fibre section extends locally. The fibre integral is computed with $h$ and any smooth volume form; the canonical-bundle factors make it independent of that volume form.

*Proof.* Zhou and Zhu prove equality in their paper published online in 2018 and in print in 2020 [115]. Optimal $L^{2/m}$ extension and generalized Siu lemmas give the required lower-limit estimate for nearby fibre norms. Their 2023 survey explains why the extremal metric uses extendable sections [116]. □

**Problem 2.6.** If $D \subset \mathbb{C}^n$ is contractible, must every smooth $\bar\partial$-closed $(p, q)$-form on $D$, $0 \leq p \leq n$ and $1 \leq q \leq n$, be $\bar\partial$-exact?

*Proof.* The answer is negative by the following elementary example. The Hartogs cross $D = (\Delta \times \frac{1}{2}\Delta) \cup (\frac{1}{2}\Delta \times \Delta)$ is star-shaped and hence contractible, but Hartogs extension shows that it is not Stein. Berg's cohomological criterion (1985) implies that not all positive-degree $\mathcal{O}$-cohomology groups of $D$ vanish [107]; the Dolbeault identification then gives a closed form without a global primitive. See also Grauert and Remmert (1979) [106]. □

**Problem 2.7.** Must a Carnot–Carathéodory conformal homeomorphism between strictly pseudoconvex CR three-manifolds be CR?

*Proof.* Tang's formulation in the 1995 problem list has an elementary counterexample [1, Problem 12.3.1]: complex conjugation on $S^3 \subset \mathbb{C}^2$ preserves the standard Carnot–Carathéodory metric but is anti-CR. Thus the stated implication is false without a convention distinguishing CR maps from anti-CR maps. □

**Problem 2.8** (Cartan–Kuranishi prolongation problem)**.** Under the regularity hypotheses of the analytic prolongation theorem, does finite prolongation of an analytic differential system produce an involutive system or detect incompatibility?

*Proof.* Kuranishi (1957) proves finite prolongation under the regularity hypotheses of the analytic theory [97]. Compatibility conditions and symbols stabilize, after which analytic existence theorems apply at ordinary data. Goldschmidt (1968) develops the precise prolongation framework for linear systems [98]. □

**Problem 2.9** (Nirenberg–Trèves conjecture)**.** For a scalar classical pseudodifferential operator of principal type, is microlocal solvability equivalent to condition $(\Psi)$ on its homogeneous principal symbol $p$: for every nonvanishing smooth factor $a$, $\operatorname{Im}(ap)$ never changes from negative to positive along an oriented null bicharacteristic of $\operatorname{Re}(ap)$? Does this give local solvability at $x_0$ when $p = 0$ implies $\partial_\xi p \neq 0$ near $T^*_{x_0}M \setminus 0$?

*Proof.* Dencker (2006) proves microlocal sufficiency of $(\Psi)$ and local solvability under the stated nontrapping hypothesis [99]. A phase-space metric and a signed-distance weight yield an estimate for the adjoint with loss of two derivatives; duality gives local solvability. □

## 2.2. **Open problems with partial results.**

**Problem 2.10.** Fix $s \geq 0$, $1 \leq q \leq n$, and a Sobolev inner product on $H^s(D)$. Develop existence and regularity criteria for the $\bar\partial$-Neumann problem defined using the adjoint of the maximal distributional $\bar\partial$ in this inner product. Separately, characterize smooth bounded pseudoconvex domains for which the usual $L^2$ operator $N_q$ extends boundedly on $H^s_{0,q}(D)$.

*Partial proof.* Christ (1996) constructs worm domains for which the usual $\bar\partial$-Neumann problem fails global $C^\infty$ regularity [34]. This excludes a theorem asserting boundedness on every Sobolev scale for every smooth pseudoconvex domain. □

**Problem 2.11.** For every bounded pseudoconvex domain $D$ with real-analytic boundary, do there exist $0 < \alpha < 1$ and $C_D > 0$ such that every closed $f \in L^\infty_{0,1}(D)$ admits $\bar\partial u = f$ with $\|u\|_{\Lambda^\alpha(\overline{D})} \leq C_D \|f\|_{L^\infty(D)}$?

*Partial proof.* Range (1990) constructs integral solution operators with a positive Hölder gain on smooth pseudoconvex finite-type domains in $\mathbb{C}^2$ [127]. Diederich, Fischer and Fornæss (1999) obtain sharp Hölder estimates on convex finite-type domains by constructing Cauchy–Fantappiè kernels from holomorphic supporting functions [128]. For comparison, Fornæss, Lee and Zhang (2011) and Khanh (2012 preprint) obtain estimates on specified infinite-type convex classes [22, 23]. □

**Problem 2.12.** For every smooth bounded convex domain $D$ and $1 < p < \infty$, does every closed $f \in L^p_{0,1}(D)$ admit $\bar\partial u = f$ with $\|u\|_{L^p(D)} \leq C_{D,p}\|f\|_{L^p(D)}$? If $D \subset \mathbb{C}^2$, does the analogous estimate hold for $p = \infty$?

*Partial proof.* Diederich, Fischer and Fornæss (1999) prove sup-norm-to-Hölder estimates on smooth convex finite-type domains [128]. Fornæss, Lee and Zhang (2011) prove sup-norm estimates for explicit infinite-type convex domains in $\mathbb{C}^2$, and Khanh's 2012 preprint gives sup-norm and $f$-Hölder estimates under quantitative hypotheses on boundary flatness [22, 23]. These constructions estimate integral kernels adapted to the defining function. □

**Problem 2.13.** Does finite D'Angelo type at a point of a smooth pseudoconvex boundary imply termination there of Kohn's multiplier algorithm for subellipticity on $(0,1)$-forms?

*Partial proof.* Nicoara (2025) proves termination for suitable Denjoy–Carleman quasianalytic classes [30], and Siu (2025) obtains an effective Kohn algorithm for special domains [31]. The arguments use additional control of multiplier ideals and their successive radicals. Nicoara's earlier 2007 preprint claiming the general smooth equivalence was subsequently withdrawn [32]. □

**Problem 2.14.** Let $\Omega \subset \mathbb{P}^n$, $n \geq 2$, be smoothly bounded and pseudoconvex. Does every closed $f \in C^\infty_{0,1}(\overline{\Omega})$ admit $u \in C^\infty(\overline{\Omega})$ with $\bar\partial u = f$?

*Partial proof.* Cao, Shaw and Wang (2004) obtain small-order Sobolev estimates in projective space; Shaw's 2023 account presents the estimate in Theorem 3.8 [51, §3.3]. A bounded strictly plurisubharmonic exhaustion supplies a positive exponent $t_0$, and weighted $L^2$ estimates give bounded solution operators on $H^s$ for $0 < s < t_0/2$. □

**Problem 2.15.** Let $X \subset \mathbb{C}^{n+p}$ be a closed pure $p$-dimensional analytic set, and give $X_{\mathrm{reg}}$ the induced metric. Describe the obstructions to $\bar\partial u = f$ in $L^2$ for closed data of bidegree $(0,q)$ in the maximal realization and $(p,q)$ in the minimal realization, $1 \leq q \leq p$. The maximal domain consists of $L^2$ forms whose distributional $\bar\partial$ is $L^2$; the minimal domain is the graph closure of compactly supported smooth forms on $X_{\mathrm{reg}}$.

*Partial proof.* Øvrelid and Vassiliadou, in their 2011 preprint, express the $L^2$ $(0,q)$-cohomology of sufficiently small deleted neighbourhoods of isolated singularities in terms of a resolution [52]. Øvrelid and Ruppenthal, in their 2012 preprint, transfer closed-range and compactness properties between the regular part and a resolution, outside specified middle degrees [53]. □

**Problem 2.16.** Let $X$ be a normal pure-dimensional closed analytic subset of a ball $B \subset \mathbb{C}^n$. If $f \in C^\infty_{0,1}(B)$ restricts to a $\bar\partial$-closed form on $X_{\mathrm{reg}}$, is there $u \in C^\infty(B)$ such that $\bar\partial(u|_{X_{\mathrm{reg}}}) = f|_{X_{\mathrm{reg}}}$?

*Partial proof.* Andersson and Samuelsson (2012) construct a Dolbeault resolution on singular complex spaces using fine sheaves of currents and Koppelman formulas [66]. This produces generalized primitives; the sheaves agree with smooth forms on the regular locus, while ambient smoothness across the singular set is a stronger requirement. □

**Problem 2.17.** Let $M = \{r = 0\} \subset \mathbb{C}^n$ be a smooth real hypersurface, with $dr \neq 0$ on $M$. Characterize the points $p \in M$ at which every germ of a smooth CR function extends holomorphically to the prescribed side $\{r > 0\}$ near $p$.

*Partial proof.* Trépreau (1986) proves that minimality of a real hypersurface gives local holomorphic extension of CR functions to one side [33]. The proof uses families of attached analytic discs and propagation of extension. The theorem supplies some side, without prescribing the sign of a chosen defining function. □

**Problem 2.18.** Is the $\bar\partial$-Neumann problem on $(0,1)$-forms globally real-analytic hypoelliptic up to the boundary of every bounded real-analytic pseudoconvex domain? Is $\Box_b$ globally real-analytic hypoelliptic on $(0,1)$-forms on every compact real-analytic pseudoconvex CR hypersurface-type manifold of real dimension at least five? In the second question, this means that $\Box_b u$ real analytic on the whole manifold forces a distributional solution $u$ to be real analytic there.

*Partial proof.* Derridj and Tartakoff (1976) prove global real-analytic regularity for the $\bar\partial$-Neumann problem in the strongly pseudoconvex setting [129]. Tartakoff (1976) proves the corresponding global result for $\Box_b$ on compact manifolds under strong pseudoconvexity [130]. Their estimates control iterated derivatives with analytic, factorial growth; Christ (1999) discusses the distinction between global and local analytic regularity [35]. □

**Problem 2.19.** Let $D \subset \mathbb{C}^n$, $n \geq 2$, have smooth boundary and let $f \in \mathcal{O}(D) \cap C^\infty(\overline{D})$. If every derivative of $f$ vanishes along a smooth nonconstant boundary curve transverse to the complex tangent spaces, must $f \equiv 0$?

*Partial proof.* Nirenberg (1974) proves uniqueness when the transverse boundary curve is real analytic [74]. Fornæss and Sibony (2001) explicitly record this case in their formulation of the smooth-curve problem [2, Question 3.15]. □

**Problem 2.20.** Let $H \subset \mathbb{B}^n$ be a smooth finite-type real hypersurface through 0. Can relatively closed analytic subvarieties $X_j \subset \mathbb{B}^n$, with $\dim_0 X_j \geq 1$, have $\limsup_{j\to\infty} X_j \subset H$? Here the limit superior consists of limits in $\mathbb{B}^n$ of sequences $x_\nu \in X_{j_\nu}$ with $j_\nu \to \infty$.

*Partial proof.* Sibony (1987) constructs plurisubharmonic barriers for pseudoconvex finite-type boundary points [79]. Applying the maximum principle to these barriers excludes an analytic cluster set contained in the hypersurface. Fornæss and Sibony (2001) record this argument for the pseudoconvex case [2, Question 3.24]. □

**Problem 2.21.** For which classes of smooth differential systems does a finite procedure involving prolongation and compatibility conditions characterize local solvability?

*Partial proof.* Kuranishi (1957) obtains finite prolongation in the regular analytic setting [97], while Dencker (2006) characterizes local solvability for scalar principal-type pseudodifferential operators by $(\Psi)$ [99]. Lewy's counterexample (1957) shows that formal compatibility of a smooth system alone is insufficient [100]. □

## 2.3. **Wide-open problems.**

**Problem 2.22.** Characterize positive measures $\mu$ for which $\bar\partial$ on smooth forms admits a densely defined closed realization in $L^2(d\mu)$ with a well-posed $\bar\partial$-Neumann problem. Determine existence and regularity estimates, beginning with the unit disc. The choice of realization is part of the problem when $\mu$ is singular.

**Problem 2.23.** Let $D \subset \mathbb{C}$ be smooth and bounded, and let $T : L^2(D) \to L^2(D)$ satisfy $\partial(Tf)/\partial\bar z = f$ and $Tf \perp A^2(D)$. For $h \in H^\infty(D)$, set $S_h f = \overline{h}\, Tf$. Does $\int_\gamma h\, dz \neq 0$ for some closed curve $\gamma \subset D$ imply that the compact operator $S_h$ has a nonzero eigenvalue?

*Remark.* D. Barrett remarks that $S_h$ has no nonzero eigenvalue when $h$ has a single-valued holomorphic primitive, and that the converse holds for annuli [1, Problem 6.1.3, 1995]. In the first case, multiplication by the exponential of the conjugate primitive reduces an eigenvector equation to a holomorphic function orthogonal to holomorphic functions.

**Problem 2.24.** For every bounded pseudoconvex $D$ and $1 < p < 2$, does each closed $(0,1)$-form $f \in L^p(D)$ admit $\bar\partial u = f$ with $\|u\|_{L^p(D)} \leq C_{D,p}\|f\|_{L^p(D)}$, without a boundary regularity assumption?

**Problem 2.25.** For domains admitting a sup-norm $\bar\partial$ estimate on closed $(0,1)$-forms, bound the optimal constant

$$C_D = \sup_{\substack{\bar\partial f = 0 \\ \|f\|_\infty \leq 1}} \inf_{\bar\partial u = f} \|u\|_\infty$$

in terms of the geometry of $D$.

*Remark.* Buzzard, Fornæss, Gavosto and Krantz remark that the constants in $L^2$ estimates depend on the diameter and dimension, offering these as a comparison for the uniform-estimate question [1, Problem 6.4.2, 1995].

**Problem 2.26.** Let $D$ be a smooth bounded pseudoconvex domain and $0 < \alpha < \beta$. Set

$$\mathcal{D}_{\alpha,\beta} = \{u \in \Lambda^{\alpha}(\overline{D}) : \bar{\partial} u \in \Lambda^{\beta}_{0,1}(\overline{D})\},$$

with its graph norm. Does $\bar{\partial} : \mathcal{D}_{\alpha,\beta} \to \Lambda^{\beta}_{0,1}(\overline{D})$ have closed range?

**Problem 2.27.** Does every bounded domain biholomorphic to $\mathbb{B}^n$ admit a sup-norm estimate for $\bar{\partial}$, without any boundary regularity assumption?

**Problem 2.28.** Is every smooth strictly pseudoconvex abstract CR manifold of real dimension five locally CR-embeddable?

*Remark.* Webster (1989) gives a proof of local embeddability for smooth strictly pseudoconvex CR manifolds of real dimension at least seven [40].

**Problem 2.29** (Complex Plateau problem)**.** Characterize compact oriented maximally complex submanifolds $M^{2p-1} \subset \mathbb{P}^n$, $1 \le p \le n$, that bound a positive holomorphic $p$-chain. Precisely, does there exist a locally finite positive integral combination $T$ of $p$-dimensional analytic subvarieties of $\mathbb{P}^n \setminus M$, of finite mass, whose boundary as a current is $dT = [M]$?

**Problem 2.30.** Fix a contact quasiconformal diffeomorphism between strictly pseudoconvex CR $(2n+1)$-manifolds, $n > 1$. Can compatible Hermitian metrics on the contact bundles minimize its distortion, measured by the ratio of the largest to the smallest singular value of the contact differential? The admissible metrics and whether minimization is pointwise or global remain to be specified in the source's research program.

**Problem 2.31.** For $D_N = \{z \in \mathbb{C}^N : \sum_{j=1}^N |z_j| < 1\}$, does every bounded continuous closed $(0,1)$-form $f$ admit $\bar{\partial} u = f$ with $\|u\|_\infty \le C\|f\|_\infty$, where $C$ is independent of $N$? Does the same hold on $\mathbb{B}^N$? Here forms carry the dual norm of the ambient $\ell^1$ norm on $D_N$, and of the Euclidean norm on $\mathbb{B}^N$, following the Banach-space convention in [67, §2].

**Problem 2.32.** Let $\phi$ be a bounded smooth strictly plurisubharmonic function on a ball $B \subset \mathbb{C}^n$, and set $\omega = i\partial\bar{\partial}\phi$. Does every smooth closed $(0,1)$-form $f$ satisfying

$$\sup_B (|f|^2_\omega + |\partial f|_\omega) < \infty$$

admit a bounded solution of $\bar{\partial} u = f$? Here both pointwise norms are induced by the Kähler metric $\omega$.

*Remark.* Fornæss and Sibony (2001) remark that Berndtsson's question is related to the corona theorem in several variables, referring to Berndtsson's weighted uniform estimates (1997) [2, Question 3.4][68].

## 3. Function theory and pluripotential theory

This category concerns holomorphic function spaces, invariant metrics, and the relation between plurisubharmonic functions and complex geometry. For a bounded domain $D$, set $A^k(\overline{D}) = \mathcal{O}(D) \cap C^k(\overline{D})$ and $A(D) = A^0(\overline{D})$; $H^\infty(D)$ denotes bounded holomorphic functions. Write $K_D(z,w)$ for the Bergman kernel, $c_D, k_D$ for the Carathéodory and Kobayashi distances, and $\gamma_D, \kappa_D$ for their infinitesimal metrics. Completeness of a hyperbolic manifold refers to its

Kobayashi distance. For a smooth bounded $D = \{r < 0\}$ with $dr \neq 0$ on $\partial D$, $H^p(D)$ consists of holomorphic $f$ for which $\sup_{0<\epsilon<\epsilon_0} \int_{\{r=-\epsilon\}} |f|^p \, d\sigma < \infty$. A peak set for $A(D)$ is a closed $E \subset \partial D$ admitting $f \in A(D)$ with $f = 1$ on $E$ and $|f| < 1$ on $\overline{D} \setminus E$. We use $dd^c = (i/\pi)\partial\bar{\partial}$ and write $\mathrm{PSH}(D)$ for the plurisubharmonic functions.

### 3.1. **Closed problems with complete solutions.**

**Problem 3.1.** Does there exist a complete hyperbolic manifold $D$ with $c_D = k_D$ that is not biholomorphic to a convex domain?

*Proof.* Agler and Young (2004) compute complex geodesics of the symmetrized bidisc and show that its Carathéodory and Kobayashi distances coincide and are complete [8]. Costara (2004) proves that this domain is not biholomorphic to a convex domain [9]. □

**Problem 3.2.** Is every bounded pseudoconvex domain with Lipschitz boundary complete for the Bergman metric?

*Proof.* Błocki and Pflug (1998) prove that every bounded hyperconvex domain is Bergman complete [14]. Bounded pseudoconvex domains with Lipschitz boundary are hyperconvex, so their theorem applies. The proof combines sublevel estimates for pluricomplex Green functions with a Bergman completeness criterion. □

**Problem 3.3** (Suita conjecture)**.** Let $\Omega$ be an open Riemann surface with negative Green function $G_\Omega(\cdot, z)$, and choose a local coordinate $w$ with $w(z) = 0$. Set

$$c_\beta(z) = \exp \lim_{\xi \to z} \big(G_\Omega(\xi, z) - \log |w(\xi)|\big).$$

Is $c_\beta(z)^2 \leq \pi K_\Omega(z)$, where $K_\Omega(z)|dw|^2$ is the diagonal Bergman kernel of holomorphic one-forms with squared norm $\frac{i}{2}\int_\Omega \eta \wedge \overline{\eta}$? When does equality hold?

*Proof.* Błocki (online 2012; print 2013) proves the inequality for bounded planar domains by sharp $L^2$ extension [111]. Guan and Zhou (2012) prove the inequality for arbitrary open Riemann surfaces admitting a Green function [109]. Guan and Zhou (2015) prove that equality at one point holds precisely when the surface is biholomorphic to a disc with a relatively closed polar set removed [110, Theorem 3.1]. Comparing the optimal extension norm with logarithmic capacity yields the inequality and identifies the equality case. □

*Remark.* Błocki's work on the bounded planar case, subsequently cited by Guan and Zhou in their 2012 paper [109, p. 754], goes a little beyond the planar inequality. He proves an $L^2$ extension estimate from the slice $\Omega \cap \{z_n = 0\}$ to a pseudoconvex domain $\Omega \subset \mathbb{C}^{n-1} \times D$, with the capacity-dependent factor $\pi/c_D(0)^2$, where $D \subset \mathbb{C}$ is a bounded domain containing 0 and $c_D$ denotes its logarithmic capacity; the planar Suita inequality follows as an application [111, Theorem 1]. His proof develops a sharpened $\bar{\partial}$ estimate and suitable auxiliary weights [111, Theorem 2 and §2].

### 3.2. **Open problems with partial results.**

**Problem 3.4.** For a smooth bounded pseudoconvex finite-type domain $D \subset \mathbb{C}^n$, characterize the peak sets for $A(D)$. In particular, is every singleton $\{p\}$, $p \in \partial D$, a peak set?

*Partial proof.* Bedford and Fornæss (1978) construct peak functions at finite-type boundary points in $\mathbb{C}^2$ [21]. Their construction produces a holomorphic function with controlled argument near the point and then converts it into a peak function. □

**Problem 3.5.** Are $c_D$ and $k_D$ complete on every smooth bounded pseudoconvex domain $D$?

*Partial proof.* Edigarian's preprint of 20 October 2026 announces a smoothly bounded pseudoconvex domain in $\mathbb{C}^3$ on which both distances are incomplete, together with Kobayashi completeness for finite D'Angelo type [15, Theorems 1.1–1.2]. The proposed counterexample uses analytic-disc chains of summable length approaching the boundary; the positive assertion uses a metric lower bound. □

**Problem 3.6** (Lu Qi-Keng problem)**.** Characterize bounded domains $D$ for which $K_D(z,w) \neq 0$ for every $z, w \in D$.

*Partial proof.* Boas, Fu and Straube (1999) compute Bergman kernels explicitly and exhibit convex domains in $\mathbb{C}^3$ whose kernels have zeros [36]. The computation disproves universal nonvanishing even within the convex class. □

**Problem 3.7.** Determine sharp boundary asymptotics of $\kappa_D(z;v)$ on smooth bounded pseudoconvex finite-type domains. After fixing local holomorphic coordinates and a defining function, which normalized asymptotic terms determine derivatives of its Levi form at a weakly pseudoconvex boundary point?

*Partial proof.* Herbort (1993), with an erratum in 1994, obtains sharp estimates for invariant differential metrics near finite-type pseudoconvex points of Levi corank one [37]. Normal forms and anisotropic scales relate the metric estimates to boundary geometry. □

**Problem 3.8.** Let $D \subset \mathbb{C}^n$ be a bounded pseudoconvex balanced domain, so that $\lambda D \subset D$ for $|\lambda| \leq 1$. Set $h(z) = \inf\{t > 0 : z/t \in D\}$, its Minkowski functional; thus $D = \{h < 1\}$ and $h(\lambda z) = |\lambda| h(z)$. Characterize completeness of each of the Carathéodory, Bergman, and Kobayashi metrics in terms of $h$.

*Partial proof.* If $h$ is continuous, $h-1$ is a negative plurisubharmonic exhaustion, so Błocki and Pflug's hyperconvexity theorem (1998) gives Bergman completeness [14]. Edigarian's preprint of 7 September 2026 announces a bounded balanced domain in $\mathbb{C}^2$ with continuous Minkowski functional and incomplete Kobayashi distance [16]. □

**Problem 3.9.** Characterize classes $\mathcal{F} \subset \mathrm{PSH}(D)$ on which the complex Monge–Ampère measure is well defined, and convergence notions on $\mathcal{F}$ for which

$$u_j \longrightarrow u \quad \Longrightarrow \quad (dd^c u_j)^n \rightharpoonup (dd^c u)^n.$$

Here $D \subset \mathbb{C}^n$ and the measures converge weakly on compact subsets of $D$.

*Partial proof.* Bedford and Taylor (1982) prove continuity for decreasing sequences of locally bounded plurisubharmonic functions with a locally bounded limit [38]. Their construction defines wedge products inductively and uses integration by parts and capacity estimates to pass to the limit. □

**Problem 3.10.** Let $D$ be bounded and Kobayashi complete, and fix $p > 0$. Do constants $C, r_0 > 0$ exist such that

$$|f(z)|^p \leq \frac{C}{\mathrm{Vol}(B_{k_D}(z,r))} \int_{B_{k_D}(z,r)} |f|^p \, dV$$

for all $f \in \mathcal{O}(D)$, $z \in D$, and $0 < r < r_0$? Here $B_{k_D}(z,r) = \{w \in D : k_D(z,w) < r\}$ and Vol is Euclidean volume.

*Partial proof.* Krantz (1995) records the strongly pseudoconvex case in every dimension and the finite-type case in $\mathbb{C}^2$ [1, Problem 13.0.3]. In these settings, comparison of Kobayashi balls with anisotropic polydiscs reduces the estimate to the submean inequality on polydiscs. □

**Problem 3.11.** Let $T$ be a positive closed current of bidimension $(p,p)$ on $\mathbb{C}^n$, $1 \le p < n-1$. Do there exist closed pure $p$-dimensional analytic varieties $V_j \subset \mathbb{C}^n$ and $c_j > 0$ with $c_j[V_j] \rightharpoonup T$ as currents? Can these be chosen so that $\operatorname{supp}[V_j]$ converges locally in the Hausdorff sense to $\operatorname{supp} T$?

*Partial proof.* Fornæss and Sibony (2001) record the hypersurface case, of bidimension $(n-1, n-1)$, as classical and cite Demailly (1999) for refinements [2, Question 3.6][69]. In that case, approximation of plurisubharmonic potentials and the Poincaré–Lelong formula give approximation by normalized divisors. □

3.3. **Wide-open problems.**

**Problem 3.12** (Corona problem)**.** Let $D = \mathbb{B}^n$ or $\Delta^n$, $n \ge 2$. If $f_1, \dots, f_m \in H^\infty(D)$ and $\sum_{j=1}^m |f_j| \ge \delta > 0$, do there exist $g_1, \dots, g_m \in H^\infty(D)$ with $\sum_{j=1}^m f_j g_j = 1$?

*Remark.* Costea, Sawyer and Wick (2010) obtain solutions in BMOA for bounded corona data on the unit ball [20].

**Problem 3.13.** For a smooth bounded pseudoconvex domain $D$, is $A^\infty(\overline{D})$ dense in $A^k(\overline{D})$ in the $C^k$ norm for every finite $k \ge 1$? If $\overline{D}$ has a Stein neighbourhood basis, is every $f \in A^\infty(\overline{D})$ uniformly approximable by functions holomorphic on neighbourhoods of $\overline{D}$?

**Problem 3.14.** Fix an identification $\mathbb{H} \simeq \mathbb{C}^2$ and a domain $\Omega \subset \mathbb{H}$. For a real orthonormal quaternionic frame $\Phi = (\Phi_0, \dots, \Phi_3)$, let $D_\Phi f = \sum_{j=0}^3 \Phi_j \partial_{x_j} f$ on $C^1(\Omega, \mathbb{H})$, with left quaternionic multiplication. Describe the nonholomorphic elements of $\ker D_\Phi$ and the ordinary holomorphic subspaces it contains, including their dependence on $\Omega$ and $\Phi$.

*Remark.* H. Shapiro and N. Vasilievsky remark that $\operatorname{Hol}(\Omega; \mathbb{C}^2) \subset \ker D_{\widehat{\Phi}}$ for $\widehat{\Phi} = (1, i, j, -k)$ [1, Problem 5.3.1, 1995].

**Problem 3.15.** Give a geometric characterization of positive probability measures $\mu$ on $\overline{\mathbb{B}^n}$ satisfying $\int f \, d\mu = f(0)$ for every $f \in A(\mathbb{B}^n)$.

**Problem 3.16.** Is every compact totally disconnected set $K \subset \mathbb{C}^n$ with finite one-dimensional Hausdorff measure polynomially convex?

**Problem 3.17.** Let $D$ be smooth, bounded and pseudoconvex, with Shilov boundary $\Sigma_D$ for $A(D)$. If an effective divisor $Z$ in $D$ satisfies $\overline{|Z|} \cap \Sigma_D = \varnothing$, must its support be contained in the zero set of some nonzero $f \in H^\infty(D)$?

*Remark.* S. G. Krantz remarks that Rudin proved the affirmative result for the polydisc [1, Problem 5.6.1, 1995].

**Problem 3.18.** For $0 < p < \infty$ and a smoothly bounded strongly pseudoconvex domain $D$, characterize the effective divisors that are zero divisors of nonzero functions in the Hardy space $H^p(D)$.

**Problem 3.19.** Does there exist a smooth bounded pseudoconvex domain $D$ for which no constant $C > 0$ satisfies

$$\kappa_D(z; v) \le C \gamma_D(z; v) \qquad (z \in D,\ v \in \mathbb{C}^n)?$$

**Problem 3.20.** At a point of a smooth pseudoconvex hypersurface, does the maximal order of contact with nonsingular holomorphic curve germs equal the commutator type obtained by using one complex tangential vector field, nonzero at that point, and its conjugate at a time? In the latter type, measure the least bracket length that produces a direction transverse to the complex tangent space, then take the supremum over the initial fields.

**Problem 3.21.** Let $D \subset \mathbb{C}^2$ have smooth pseudoconvex finite-type boundary. For each $p \in \partial D$, does a neighbourhood $U$ exist such that $D \cap U$ is biholomorphic to a convex domain by a map extending continuously to $\partial D \cap U$?

**Problem 3.22.** Let $V$ be an irreducible analytic germ at $0 \in \mathbb{C}^n$ and $E = V \cap \mathbb{R}^n \cap B_\epsilon$. Classify the germs for which the extremal function $U_E$ satisfies $U_E(z) \le A|\operatorname{Im} z|$ near 0. The printed ambient-ball envelope uses functions between 0 and 1 vanishing on $E$, but its omitted admissible function class must be fixed without silently replacing it by an intrinsic envelope on $V$.

**Problem 3.23.** For every smooth bounded strongly pseudoconvex domain $D$, does there exist an integer $N_D$ such that, for every $z, w \in D$, $k_D(z, w)$ is attained by a chain of at most $N_D$ analytic discs? Attainment means that the sum of the Poincaré distances between the paired preimages equals $k_D(z, w)$.

*Remark.* S. G. Krantz remarks that $N_D = 1$ for convex domains [1, Prize Problem 1, 1995].

**Problem 3.24.** Let $p$ be a finite-type real-analytic boundary point of a domain $\Omega \subset \mathbb{C}^n$, and suppose $p$ is not in the image of its envelope of holomorphy under the natural projection to $\mathbb{C}^n$. Do there exist $\epsilon, C > 0$ and a neighbourhood $U$ of $p$ such that

$$\kappa_\Omega(q; v) \ge \frac{C\|v\|}{\|p - q\|^\epsilon} \qquad (q \in \Omega \cap U,\ v \in \mathbb{C}^n)?$$

*Remark.* Fornæss and Sibony (2001) remark that such a lower bound would be useful for proving continuous boundary extension of proper holomorphic maps [2, Question 3.22].

## 4. Holomorphic mappings and hyperbolicity

Mapping rigidity asks how algebraic, intrinsic, or boundary geometry constrains holomorphic maps. A complex manifold is *Kobayashi hyperbolic* when its Kobayashi pseudodistance separates points; its infinitesimal pseudometric is denoted by $\kappa_X$. We write $\operatorname{Aut}(D)$ for the holomorphic automorphism group of a domain $D$. A *general* hypersurface is chosen outside a proper algebraic subset of its parameter space; *very general* allows a countable union of proper algebraic subsets.

### 4.1. Closed problems with complete solutions.

**Problem 4.1** (Kobayashi conjecture)**.** For every $n \ge 2$, does there exist $d_0(n)$ such that a general hypersurface $X_d \subset \mathbb{P}^n$ of degree $d$ is Kobayashi hyperbolic whenever $d \ge d_0(n)$?

*Proof.* Brotbek (2017) proved the assertion [87]. Wronskians produce global jet differentials on suitable families of high-degree hypersurfaces. Positivity and base-locus control force an entire curve to satisfy sufficiently many differential equations, excluding nonconstant entire curves on a general member. Brody's criterion then gives Kobayashi hyperbolicity because the hypersurface is compact. □

### 4.2. Open problems with partial results.

**Problem 4.2.** Determine the minimum of $\deg F := \max(\deg F_1, \deg F_2)$ among noninjective polynomial maps $F : \mathbb{R}^2 \to \mathbb{R}^2$ satisfying $\det DF(x) \neq 0$ for all $x \in \mathbb{R}^2$.

*Partial proof.* Fernandes (2022) constructed a polynomial local diffeomorphism with coordinate degrees 10 and 15 which is not injective [27]. Its Jacobian never vanishes, whereas distinct points have the same image. Therefore the least possible value of $\max(\deg F_1, \deg F_2)$ is at most 15. □

**Problem 4.3.** For each $n \geq 1$ and $d \geq 2n+1$, is $\mathbb{P}^n \setminus D$ Kobayashi hyperbolic for a general degree-$d$ hypersurface $D \subset \mathbb{P}^n$?

*Partial proof.* Brotbek and Deng (2019) proved hyperbolicity of complements of general hypersurfaces of sufficiently large degree, using logarithmic Wronskians and logarithmic jet differentials [56]. Hou, Huynh, Merker, and Xie (September 2026 preprint revision) obtain the bound $d \geq 12$ for complements of very generic plane curves by vanishing results for negatively twisted invariant 2-jet differentials [57]. The latter is a preprint result; the proposed threshold in the plane is $d = 5$. □

**Problem 4.4.** Classify smoothly bounded domains $D \Subset \mathbb{C}^n$ with noncompact $\mathrm{Aut}(D)$. If $D$ is pseudoconvex, must $\mathrm{Aut}(D)$ contain a subgroup topologically isomorphic to $(\mathbb{R}, +)$?

*Partial proof.* Bedford and Pinchuk (1988; English translation 1989) proved that a bounded pseudoconvex domain in $\mathbb{C}^2$ with real-analytic boundary and noncompact automorphism group is biholomorphic to

$$E_m = \{(z,w) \in \mathbb{C}^2 : |z|^2 + |w|^{2m} < 1\}, \qquad m \geq 1$$

for an integer $m$ [39]. They rescale near an orbit-accumulation point and classify the resulting polynomial model. The maps

$$\Phi_t(z,w) = \left( \frac{z\cosh t + \sinh t}{z\sinh t + \cosh t}, \frac{w}{(z\sinh t + \cosh t)^{1/m}} \right), \qquad t \in \mathbb{R},$$

form a subgroup of $\mathrm{Aut}(E_m)$: direct substitution gives $\Phi_t(E_m) = E_m$ and $\Phi_s \circ \Phi_t = \Phi_{s+t}$. The holomorphic root is chosen positive at $z = 0$. Since $\Phi_t(0,0) = (\tanh t, 0)$, this subgroup is topologically isomorphic to $(\mathbb{R}, +)$ and noncompact. Thus the subgroup question also has an affirmative answer in this class. □

**Problem 4.5.** Classify irreducible analytic curves $\Gamma \subset \Delta \times \Delta$ for which both coordinate projections $\pi_j : \Gamma \to \Delta$, $j = 1, 2$, are proper and finite-to-one.

*Partial proof.* For the disc-normalization case of the question recorded by Buzzard, Fornæss, Gavosto, and Krantz (1995) [1, Problem 9.3.1], an elementary argument applies. Let the normalization be $\Delta$, and let its two proper projections be $f, g : \Delta \to \Delta$. A proper disc map has finitely many zeros and modulus tending uniformly to 1 at the boundary. Dividing $f$ by the finite Blaschke product with its zeros leaves a zero-free holomorphic function of boundary modulus 1; the maximum principle for it and its reciprocal makes it a unimodular constant. The same holds for $g$. Thus the correspondence is parametrized by a pair of finite Blaschke products. □

**Problem 4.6.** Classify pairs $(\Sigma, X)$, where $\Sigma \subset \mathbb{C}^n$ is a real-analytic finite-type hypersurface and $X$ is a holomorphic vector field near $\Sigma$ satisfying $\mathrm{Re}\, X|_\Sigma \in T\Sigma$.

*Partial proof.* Bedford and Pinchuk (1994; English translation 1995) studied holomorphic vector fields tangent to the boundaries of weighted-homogeneous model domains and applied the resulting restrictions to classify bounded smooth convex finite-type domains with noncompact automorphism groups [77]. The method decomposes a tangent field into homogeneous terms and uses tangency to constrain the defining polynomial. □

**Problem 4.7.** Let $F : D \to D'$ be a biholomorphism between bounded domains in $\mathbb{C}^n$. If $\partial D$ and $\partial D'$ are real analytic, must $F$ extend continuously to $\overline{D}$? Does the same hold for $C^\infty$ boundaries? If $D$ and $D'$ are smoothly bounded and pseudoconvex, must $F \in C^\infty(\overline{D})$?

*Partial proof.* Fefferman (1974) proved that a biholomorphism between bounded $C^\infty$ strongly pseudoconvex domains extends to a $C^\infty$ diffeomorphism of their closures [78]. The transformation law and boundary asymptotics of the Bergman kernel control the map and its derivatives at the boundary. □

### 4.3. **Wide-open problems.**

**Problem 4.8** (Jacobian conjecture)**.** Let $F \in \mathbb{C}[z_1, z_2]^2$ satisfy $\det DF \equiv c \in \mathbb{C}^*$. Must $F : \mathbb{C}^2 \to \mathbb{C}^2$ be a polynomial automorphism?

*Remark.* T. Tao (2026) explains the counterexample in $\mathbb{C}^3$ announced by L. Alpöge following work with Claude Fable on a question suggested by A. Mathew [139, 140]. The polynomial map has $\det DF \equiv -2$ and is noninjective; adjoining identity coordinates gives counterexamples in every dimension $n \geq 3$, while the plane case stated above remains open. A. F. Ramos, D. B. Hulak, and R. J. G. Barretto de Queiroz (2026) independently verify the determinant and noninjectivity in Isabelle/HOL [140].

**Problem 4.9.** Let $D \subset \mathbb{P}^n$ be a hypersurface of degree at least $n+2$ with only normal-crossing singularities. Does there exist a hypersurface $E \subset \mathbb{P}^n$ such that

$$\kappa_{\mathbb{P}^n \setminus D}(x; v) > 0 \quad \text{for every } x \in \mathbb{P}^n \setminus (D \cup E), \quad 0 \neq v \in T_x\mathbb{P}^n?$$

**Problem 4.10.** Let $D \Subset \mathbb{C}^n$, $n > 1$, have smooth boundary. Must every proper holomorphic map $F : D \to D$ be biholomorphic?

**Problem 4.11.** Let $\Sigma, \Sigma' \subset \mathbb{C}^n$ be real-analytic hypersurfaces, with $\Sigma'$ strictly pseudoconvex and either nonspherical or simply connected. Let $p \in \Sigma$ be strictly pseudoconvex and let $f : (\mathbb{C}^n, p) \to \mathbb{C}^n$ be a holomorphic germ with $f(\Sigma) \subset \Sigma'$. Must $f$ admit analytic continuation along every path in $\Sigma$ starting at $p$?

*Remark.* J. E. Fornæss and N. Sibony (2001) attribute the question to S. Pinchuk and refer to special cases studied by Pinchuk, Shafikov, and Bedford–Pinchuk [2, Question 3.18]. In particular, Shafikov's continuation results appeared in 2000 [76].

## 5. Complex dynamics and foliations

Iteration connects the geometry of Fatou components with invariant currents, hyperbolicity, and holomorphic foliations. The questions below concern polynomial automorphisms, projective endomorphisms, and several related conservative systems.

For a holomorphic endomorphism $f : \mathbb{P}^k \to \mathbb{P}^k$ of algebraic degree $d \geq 2$, write

$$T = \lim_{m\to\infty} d^{-m}(f^m)^*\omega_{\mathrm{FS}}, \qquad \mu = T^k, \qquad J_j = \operatorname{supp}(T^j),$$

where $\int_{\mathbb{P}^k} \omega_{\mathrm{FS}}^k = 1$. The Fatou set is the normality locus of $(f^m)$, and $\Omega(f)$ denotes the nonwandering set. A Fatou component is *wandering* if its forward images are not eventually periodic. On a Siegel or rotation domain, the iterates of a return map have compact closure in the automorphism group. Write $\mathrm{End}_d(\mathbb{P}^k)$ for the coefficient parameter space of degree-$d$ endomorphisms; Lebesgue-null subsets are defined in its local coordinates.

For a generalized Hénon map $f$ of $\mathbb{C}^2$, let $K^\pm$ be the sets of bounded forward and backward orbits, $J^\pm = \partial K^\pm$, $J = J^+ \cap J^-$, and $J^* = \operatorname{supp} \mu_H$. With $d = \deg f > 1$, set

$$G^\pm(z) = \lim_{m\to\infty} d^{-m} \log^+ \|f^{\pm m}(z)\|, \qquad T^\pm = dd^c G^\pm, \qquad \mu_H = T^+ \wedge T^-,$$

where $dd^c = (i/\pi)\partial\bar\partial$. An attracting set is a compact set $A = \bigcap_{n\geq 0} f^n(U)$ for an open trapping region $f(U) \Subset U$; it is an attractor if it has a dense forward orbit, and is nontrivial if it is neither

a periodic orbit nor the whole space. A real hypersurface is *Levi-flat* if its complex tangent bundle is integrable. Hausdorff dimension is denoted by $\dim_H$, with $\dim_H \mu = \inf\{\dim_H E : \mu(E) = 1\}$ over Borel sets $E$.

### 5.1. **Closed problems with complete solutions.**

**Problem 5.1** (Bedford's stable-manifold conjecture)**.** Let $f : X \to X$ be a holomorphic automorphism of a complex manifold, uniformly hyperbolic on a compact invariant set $\Lambda$. For $p \in \Lambda$ with $\dim_{\mathbb{C}} E^s_p = r$, is $W^s(p)$ biholomorphic to $\mathbb{C}^r$?

*Proof.* S. Bera and K. Verma (2024) prove that every uniformly attracting non-autonomous basin in $\mathbb{C}^r$ is biholomorphic to $\mathbb{C}^r$ [44]. Local stable charts along the orbit identify the stable manifold with such a basin; uniform hyperbolicity supplies uniform contraction bounds. Applying their basin theorem gives the required biholomorphism. The theorem is obtained by conjugating the sequence to controlled polynomial models and constructing a convergent sequence of biholomorphic approximations. □

**Problem 5.2.** Does there exist a non-Runge domain $D \subsetneq \mathbb{C}^2$ biholomorphic to $\mathbb{C}^2$?

*Proof.* E. F. Wold (2008) constructs a Fatou–Bieberbach embedding by successive automorphism approximations, while arranging that its image $D$ contains a compact set $K$ with $\widehat{K} \not\subset D$ [5]. Here $\widehat{K}$ denotes the polynomial hull in $\mathbb{C}^2$. A Runge domain would contain the polynomial hull of each of its compact subsets, so this $D$ is not Runge. □

**Problem 5.3.** Do wandering Fatou components occur for polynomial Hénon maps of $\mathbb{C}^2$, and for holomorphic endomorphisms of $\mathbb{P}^2$ of degree at least two?

*Proof.* P. Berger and S. Biebler (2023) construct degree-six polynomial automorphisms with wandering Fatou components using homoclinic tangencies and renormalization [10]. M. Astorg, X. Buff, R. Dujardin, H. Peters, and J. Raissy (2016) construct skew products

$$F(z,w) = (p(z) + \tfrac{\pi^2}{4}w, q(w)), \qquad p(z) = z + z^2 + O(z^3), \quad q(w) = w - w^2 + O(w^3),$$

whose parabolic transition has an attracting Lavaurs fixed point [11]. Increasing return times yield a wandering component. Their choices permit $\deg p = \deg q$, so $F$ extends holomorphically to $\mathbb{P}^2$. □

**Problem 5.4.** Does every holomorphic endomorphism $f : \mathbb{P}^k \to \mathbb{P}^k$ of degree $d \geq 2$ have repelling periodic points? Are all Lyapunov exponents of $\mu$ positive?

*Proof.* J.-Y. Briend and J. Duval (1999) prove that all equilibrium Lyapunov exponents satisfy $\lambda_j \geq \frac{1}{2}\log d$ [12]. Pluripotential estimates provide many inverse branches with exponential contraction. Fixed points of these inverse branches give repelling periodic points of $f$, and counting the branches yields equidistribution toward $\mu$. □

**Problem 5.5.** Is $J_k$ the closure of all repelling periodic points of a projective endomorphism?

*Proof.* J. E. Fornæss and N. Sibony (2001) construct projective endomorphisms with isolated repelling periodic orbits outside $\operatorname{supp}\mu$ by perturbing generalized Hénon maps [13]. Such an orbit belongs to the closure of the repelling periodic points but not to $J_k$, disproving the equality. The equidistribution theorem of J.-Y. Briend and J. Duval (1999) allows exceptional repelling orbits whose normalized counting mass tends to zero [12]. □

**Problem 5.6.** Can the basin $B$ of an attracting periodic point of a generalized Hénon map have $C^1$ boundary?

*Proof.* E. Bedford and K. Kim (2017) prove that $J^+$ is not a $C^1$ manifold, even allowing a manifold boundary [41]. The basin-boundary identity of E. Bedford and J. Smillie (1991), recalled in that paper, gives $\partial B = J^+$. Thus $\partial B$ cannot be $C^1$. The nonsmoothness proof derives a Levi foliation and restrictions on unstable multipliers, then contradicts those restrictions using the algebra of periodic points. □

**Problem 5.7.** Which holomorphic endomorphisms of $\mathbb{P}^k$ have an equilibrium measure absolutely continuous with respect to volume?

*Proof.* F. Berteloot and C. Dupont (2005) prove that absolute continuity of $\mu$ forces $f$ to be a Lattès endomorphism [46]. They linearize inverse dynamics at typical points, derive regularity of the Green current, and obtain the torus-quotient structure. Conversely, for a Lattès endomorphism, $\mu$ is the pushforward of Haar measure under the finite holomorphic quotient from a complex torus; this measure is absolutely continuous with respect to projective volume. □

**Problem 5.8.** Let $f \in \mathrm{Aut}(\mathbb{C}^k)$ have an attracting fixed point $p$ with basin $B$, and set

$$B_\delta^* = \{z \in B : \operatorname{dist}(z, \partial B) \geq \delta,\ \|z - p\| \geq \delta\}.$$

For each $\delta > 0$, must there be $N = N(f, p, \delta)$ such that no orbit segment $z, f(z), \dots, f^N(z)$ is contained in $B_\delta^*$?

*Proof.* The unrestricted affine formulation recorded by J. E. Fornæss and N. Sibony (2001) admits a direct counterexample [2, Question 2.14]. Take $f(z) = z/2$, $p = 0$, and $B = \mathbb{C}^k$. For any proposed bound $N$, choose $\|z\| > 2^N\delta$. Then $\|f^n(z)\| > \delta$ for $0 \leq n \leq N$, while the boundary-distance condition is vacuous because $\partial B = \varnothing$. Thus no bound independent of the initial point exists. □

**Problem 5.9.** For the standard map

$$p_{n+1} = p_n + K \sin\theta_n, \qquad \theta_{n+1} = \theta_n + p_{n+1} \pmod{2\pi},$$

does $K > 1$ imply that Lebesgue-almost every initial point in $S^1 \times \mathbb{R}$ has unbounded forward momentum?

*Proof.* At $(\theta, p) = (\pi, 0)$ the derivative has determinant one and trace $2 - K$, so the fixed point is elliptic for $0 < K < 4$. For suitable nonresonant $K \in (1, 4)$ its Birkhoff twist is nonzero. J. Moser's invariant-curve theorem (1962) then gives invariant Jordan curves surrounding the fixed point [131]; this standard-map mechanism is discussed by J. D. Meiss (1992) [48]. The interior of such a curve is invariant and has positive area and bounded momentum, contradicting the asserted almost-everywhere escape. □

## 5.2. **Open problems with partial results.**

**Problem 5.10.** Classify periodic Fatou components of generalized Hénon maps and projective endomorphisms. Can a Hénon map have a Herman-ring basin? For $f \in \mathrm{End}_d(\mathbb{P}^k)$ and an invariant Fatou component $U$, does $\operatorname{dist}(f^n(z), \partial U) \to 0$ locally uniformly on $U$ force $\partial U$ to contain a parabolic fixed point?

*Partial proof.* J. E. Fornæss and N. Sibony (1995) classify recurrent domains in their two-dimensional setting by studying holomorphic limits of iterates and the rank of the resulting retractions [59]. M. Lyubich and H. Peters (2014) treat nonrecurrent periodic components of Hénon maps with $|\operatorname{Jac} f| < d^{-2}$, proving that they are semi-parabolic basins [29]. Their analysis of limit maps and boundary convergence supplies the missing nonrecurrent case under

this dissipativity hypothesis; the recurrent possibilities still include the Herman-ring alternative. □

**Problem 5.11.** For a generalized Hénon map, determine the inclusions among $J$, $J^*$, and $\Omega(f)$. When does hyperbolicity on $J^*$ imply $J = J^*$ and hyperbolicity on $\Omega(f)$?

*Partial proof.* L. Guerini and H. Peters (2018) prove $J = J^*$ when $|\operatorname{Jac} f| < d^{-2}$ and $f$ is hyperbolic on $J^*$ [49]. They control stable and unstable parametrizations, using substantial dissipativity to exclude additional points of $J \setminus J^*$. Their paper also treats quasi-hyperbolicity and dominated splitting under its stated hypotheses. □

**Problem 5.12.** For a projective endomorphism, set $J_0 = \Omega(f|_{J_1})$. Determine $\dim_H \mu$, $\dim_H J_1$, $\dim_H J_0$, and $\dim_H(J_1 \setminus J_0)$, and describe $J_0 \setminus J_k$. For a polynomial endomorphism of $\mathbb{C}^k$ extending holomorphically to $\mathbb{P}^k$, is $\operatorname{supp}\mu$ relatively open in $\Omega(f|_{\mathbb{C}^k})$?

*Partial proof.* I. Binder and L. DeMarco (2003) prove $\dim_H \mu \le 2k-1$ for regular polynomial endomorphisms of $\mathbb{C}^k$ [50]. Their inverse-branch and volume estimates produce a full-measure set with this Hausdorff-dimension bound. J. E. Fornæss and N. Sibony (2001) also record relative openness of $\operatorname{supp}\mu$ in $\Omega(f)$ when that support is hyperbolic [2, Question 2.17]. □

**Problem 5.13.** For fixed $d \ge 2$ and $k \ge 2$, can the locus of maps in $\operatorname{End}_d(\mathbb{P}^k)$ with infinitely many attracting cycles have nonempty interior or positive Lebesgue measure? In particular, is this locus null when $k = 2$?

*Partial proof.* G. T. Buzzard (1997) constructs holomorphic maps of two variables with infinitely many attracting cycles by a complex Newhouse construction [58]. Persistent homoclinic tangencies permit successive perturbations creating new attracting cycles while retaining previous ones. Nested parameter choices then give infinitely many sinks. J. E. Fornæss and N. Sibony (2001) discuss this construction in their projective parameter-space question [2, Question 2.1]. This is an existence result in the holomorphic two-variable setting; no conclusion about the measure of the stated projective locus is inferred. □

**Problem 5.14.** Classify holomorphic germs $f : (\mathbb{C}^k, 0) \to (\mathbb{C}^k, 0)$ up to local holomorphic conjugacy. In dimension two, determine the dynamics when

$$f(z,w) = (e^{i\theta}z, e^{i\psi}w) + O(\|(z,w)\|^2),$$

including resonant and non-linearizable cases.

*Partial proof.* For the tangent-to-identity special case, M. Hakim (1998) constructs parabolic curves associated with nondegenerate characteristic directions [65]. Blowing up the fixed point separates these directions; invariant sectors and a functional equation then produce holomorphic curves on which the iterates converge to the origin tangentially to the prescribed direction. □

**Problem 5.15.** For fixed degree $d \ge 2$, construct and describe families of endomorphisms of $\mathbb{P}^2$ whose critical hypersurface components are all preperiodic. Classify such endomorphisms of $\mathbb{P}^3$ up to projective conjugacy.

*Partial proof.* The coordinate power maps $[z_0 : \cdots : z_k] \mapsto [z_0^d : \cdots : z_k^d]$ give examples: each critical coordinate hyperplane is invariant. For the stronger class of maps that are 2-critically finite in his sense, M. Jonsson (1998) proves that $\mathbb{P}^2$ has no proper nonempty closed backward-invariant subset [62]. Applying this to $J_2 = \operatorname{supp}\mu$ gives $J_2 = \mathbb{P}^2$; he also proves density of repelling periodic points. His argument uses hyperbolicity of postcritical complements and normal inverse branches. The coordinate power maps do not satisfy his stronger hypothesis. □

**Problem 5.16.** For a kicked-rotor unitary operator $U$ on $L^2(S^1, d\theta/2\pi)$, find nonresonance hypotheses ensuring that every $\psi$ with $\|\psi\|_2 = 1$ admits $M = M(U, \psi) < \infty$ such that

$$\sum_{|j| \le M} |\widehat{U^n \psi}(j)|^2 > \tfrac{1}{2} \qquad (n \ge 0).$$

*Partial proof.* J. Bourgain (2002) proves pure-point localization in the small-coupling parameter regimes specified in his kicked-rotor theorem, using Green-function estimates and multiscale analysis [60]. For any such operator, approximate $\psi$ by a finite linear combination of eigenvectors. Its orbit is contained in a finite-dimensional compact torus, and the approximation error remains unchanged under every $U^n$. Hence the orbit of $\psi$ is precompact in $L^2$. Fourier projections $P_M$ converge uniformly to the identity on its compact closure, so $\sup_n \|(I - P_M)U^n\psi\|_2^2 < 1/2$ for some $M$, proving the asserted bound in these localization regimes. □

**Problem 5.17.** Characterize degree-$d$ endomorphisms of $\mathbb{P}^k$ for which $J_k = \mathbb{P}^k$, and determine the Hausdorff dimension of their parameter locus.

*Partial proof.* For a Lattès endomorphism, F. Berteloot and C. Dupont (2005) identify the equilibrium measure with the pushforward of Haar measure on the covering complex torus [46]. Every nonempty open subset of $\mathbb{P}^k$ has nonempty open inverse image under the quotient map, hence positive Haar measure. Therefore $\operatorname{supp} \mu = \mathbb{P}^k$. □

**Problem 5.18.** For some fixed $d \ge 2$, does the set of maps in $\mathrm{End}_d(\mathbb{P}^2)$ admitting a nontrivial attractor have positive Lebesgue measure? Determine the possible Hausdorff dimensions of these attractors.

*Partial proof.* J. Taflin (2021) constructs open sets of projective endomorphisms with attracting sets of nonempty interior, and constructs a transitive attractor with nonempty interior [63]. Saddle blenders give robust intersections with families of holomorphic disks; trapping regions convert these intersections into attracting sets. Any example with nonempty interior in $\mathbb{P}^2$ has Hausdorff dimension four. The open-family statement for attracting sets and the transitive example are separate conclusions of the construction. □

**Problem 5.19.** Can $\mathbb{P}^2$ contain a compact set laminated by nonsingular holomorphic curves that is not a compact complex curve? For a singular holomorphic foliation $\mathcal{F}$ of $\mathbb{P}^2$, must every leaf $L$ satisfy $\overline{L} \cap \operatorname{Sing} \mathcal{F} \neq \varnothing$?

*Partial proof.* Assume additionally that there are no compact leaves, that the lamination extends to a $C^1$ foliation near its compact support, and that it carries a nonzero locally finite holonomy-invariant transverse measure. The measure is nonatomic, so the intersection theorem of S. Hurder and Y. Mitsumatsu (1991) gives $[S]^2 = 0$ for its positive closed directed current $S$ [132]. But on $\mathbb{P}^2$, $[S] = a[\omega_{\mathrm{FS}}]$ with $a > 0$, so $[S]^2 = a^2 > 0$. This excludes the measured $C^1$ case, not arbitrary transversely continuous laminations. J. E. Fornæss and N. Sibony (2001) record this obstruction [2, Question 2.29]; R. Shafikov (2025) discusses the connection with exceptional minimal sets [55]. □

**Problem 5.20.** For a hyperbolic solenoid with complex stable dimension two, construct a nonzero positive closed current directed by the stable lamination. When is such a current, after normalization, canonical?

*Partial proof.* Assume that the stable lamination has closed support in an ambient complex $n$-manifold and carries a nonzero locally finite holonomy-invariant transverse measure $\nu$. The

construction of D. Ruelle and D. Sullivan (1975) defines, in each flow box with holomorphic plaques $L_t$,

$$S = \int [L_t]\, d\nu(t).$$

The flow-box construction has locally finite mass, holonomy invariance makes it independent of charts, and Stokes' theorem gives $dS = 0$ [133]. Thus $S$ is a nonzero positive current of bidimension $(2,2)$. A uniquely normalized transverse measure gives a unique current by this construction. Separately, S. Bera and K. Verma (2024) prove the stable-leaf uniformization requested in the original question for uniformly hyperbolic automorphisms [44]; see [2, Question 2.28]. □

5.3. **Wide-open problems.**

**Problem 5.21.** Does $\mathbb{P}^2$ contain a compact nonsingular $C^2$ Levi-flat hypersurface? More generally, can a compact topological hypersurface in $\mathbb{P}^2$ be locally laminated by disjoint holomorphic discs?

**Problem 5.22.** Does some conservative quadratic Hénon map

$$F_c(z,w) = (z^2 + c - w, z)$$

have a Siegel domain containing a periodic point? More generally, can a nonlinear entire $g$, with $g(0) = 0$, make $(z,w) \mapsto (g(z) - w, z)$ have a Siegel domain at the origin? For a Hénon germ with multipliers $e^{i\theta}, e^{-i\theta}$, find arithmetic conditions ensuring two invariant Siegel discs tangent to its eigenlines and contained in $J$.

*Remark.* J. E. Fornæss and N. Sibony (2001), recording M. Herman's question, remark that the reciprocal multipliers $e^{i\theta}$ and $e^{-i\theta}$ are resonant and that formal linearization imposes infinitely many compatibility conditions [2, Question 2.7].

**Problem 5.23.** Must every holomorphic endomorphism of $\mathbb{P}^2$ of degree at least two have a repelling fixed point?

**Problem 5.24.** Let $f \in \mathrm{End}_d(\mathbb{P}^2)$ and let $K$ be a nonempty compact set with $f^{-1}(K) = K = f(K)$. Does

$$K \cap \overline{\bigcup_{n\geq 1} f^n(C_f)} = \varnothing$$

imply uniform hyperbolicity on $K$, where $C_f$ is the critical set? Does avoidance of the unclosed postcritical set suffice?

**Problem 5.25.** Can a Siegel domain of a Hénon map, or of an arbitrary automorphism of $\mathbb{C}^2$, have strongly pseudoconvex boundary? What restrictions follow if the boundary of a Hénon Fatou component has a smooth relatively open hypersurface piece?

*Remark.* J. E. Fornæss and N. Sibony (2001) contrast this question with R. Pérez-Marco's one-variable Siegel discs with smooth boundary, citing that construction as a preprint [2, Question 2.8].

**Problem 5.26.** Which Reinhardt domains occur, up to biholomorphism, as Siegel domains of automorphisms of $\mathbb{C}^k$? Which occur for endomorphisms of $\mathbb{P}^k$?

**Problem 5.27.** For a projective endomorphism $f$, call $A$ a Siegel analytic set if it is closed analytic in some open set $V$ and $f^{n_j}|_A \to \mathrm{id}_A$ locally uniformly for a sequence $n_j \to \infty$. Is $\Omega(f)$ the closure of the union of periodic points and all Siegel analytic sets? Can the analogous assertion fail for entire maps of $\mathbb{C}^2$?

**Problem 5.28.** Is the closure of the repelling periodic points of a projective endomorphism relatively open in $\Omega(f)$? For an entire map of $\mathbb{C}^k$, can a repelling periodic point be a limit of saddle periodic points, or of points whose forward orbits have both bounded and unbounded subsequences?

*Remark.* J. E. Fornæss and N. Sibony (2001) remark that saddle periodic points cannot accumulate at a repelling periodic point of a biholomorphism: for a suitable iterate of the inverse map, the repelling point is attracting and has a neighborhood containing no other periodic orbit [2, Question 2.16].

**Problem 5.29.** Characterize the projective endomorphisms whose Green current $T$ spans an extremal ray in the cone of positive closed $(1,1)$-currents on $\mathbb{P}^k$.

*Remark.* J. E. Fornæss and N. Sibony (2001) record extremality for homogeneous polynomial examples and attribute to M. Jonsson the observation that the Green current of the Ueda-type map $[z : w : t] \mapsto [z^2 : w^2 - 2zt : t^2]$ is not extremal [2, Question 2.19].

**Problem 5.30.** Writing $f^n = (g_n, h_n)$, can a Hénon map admit $p \in J^*$ such that

$$\inf_{n \geq 0} |g_n(p) - g_n(q)| = 0 \qquad \text{for every } q \in J^*?$$

*Remark.* J. E. Fornæss and N. Sibony (2001) remark that replacing the first-coordinate distance by $\|f^n(p) - f^n(q)\|$ makes the asserted property impossible, and attribute this obstruction to the then-preprint of A. M. Bonifant and J. E. Fornæss on collision attractors [2, Question 2.23].

**Problem 5.31.** In the Whitney fine topology on holomorphic symplectomorphisms of $\mathbb{C}^{2k}$ preserving $\mathbb{R}^{2k}$, do maps for which

$$K_f^{\mathbb{R}} = \{x \in \mathbb{R}^{2k} : \sup_{n \geq 0} \|f^n(x)\| < \infty\}$$

has empty interior in $\mathbb{R}^{2k}$ form a dense $G_\delta$?

*Remark.* J. E. Fornæss and N. Sibony (2001) note that the real-preserving holomorphic symplectomorphisms form a Baire space in the Whitney fine topology used in the question [2, Question 2.25].

**Problem 5.32.** Fix $N \geq 2$ and $k \geq 2$. Let $\Phi_H^t(z)$ be the maximal real-time flow of the holomorphic Hamiltonian vector field $X_H$, defined for $t \in I_H(z) \subset \mathbb{R}$. For Lebesgue-almost every polynomial $H$ of degree at most $N$ and almost every $z \in \mathbb{C}^{2k}$, is

$$\sup_{t \in I_H(z)} \|\Phi_H^t(z)\| = \infty?$$

**Problem 5.33.** Classify smooth compact Levi-flat hypersurfaces in complex tori.

**Problem 5.34.** For holomorphic self-maps of a specified complex Banach or Hilbert space, formulate a notion of chaotic dynamics compatible with finite-dimensional holomorphic dynamics and with quantum localization.

## 6. Complex geometry and algebraic cycles

The questions in this section concern the passage from analytic positivity to algebraic structure, and the behaviour of complex structures under deformation. Unless stated otherwise, $X$ is a smooth projective complex variety of dimension $n$, $K_X = \bigwedge^n T_X^*$ is its canonical bundle, and $P_m(X) = h^0(X, K_X^{\otimes m})$ is its $m$th plurigenus. We write $mK_X$ for $K_X^{\otimes m}$. A line bundle is positive if it admits a smooth Hermitian metric with positive curvature, and pseudoeffective if it admits a singular Hermitian metric with nonnegative curvature current. A variety is uniruled if a general point lies on a rational curve.

### 6.1. **Closed problems with complete solutions.**

**Problem 6.1** (Kodaira embedding theorem)**.** Is a compact complex manifold $M$ projective if and only if there exist a holomorphic line bundle $L \to M$ and a smooth Hermitian metric $h$ with $i\Theta_h(L) > 0$?

*Proof.* Kodaira (1954) proved the assertion [108]. Positivity gives enough sections of sufficiently large tensor powers to separate points and tangent directions, hence a projective embedding. Conversely, the restriction of the Fubini–Study metric makes the hyperplane bundle positive. □

*Remark.* In their 2026 preprint, Du and Xie construct an ample rank-two bundle on an abelian surface with no smooth Griffiths-semipositive Hermitian metric [148, Theorem 1.1]; they disclose generative-AI assistance and state that they independently verified the final arguments. OpenAI reports a different family on $\mathbb{P}^1 \times \mathbb{P}^1$ excluding smooth strictly Griffiths-positive metrics [149, Theorem 1.1]. Liu and Wan remark that excluding Hermitian metrics does not by itself exclude Finsler metrics; their Hirzebruch-surface examples exclude smooth strongly pseudoconvex Finsler metrics of semipositive Kobayashi curvature, with ChatGPT acknowledged as an auxiliary tool [150, Theorem 1.1 and AI Declaration]. These surface examples do not settle the higher-dimensional splitting question in Problem 6.6.

**Problem 6.2.** Is the canonical ring

$$R(X, K_X) = \bigoplus_{m \geq 0} H^0(X, mK_X)$$

finitely generated as a $\mathbb{C}$-algebra?

*Proof.* Birkar, Cascini, Hacon, and McKernan (2010) proved finite generation [94]. Their minimal model program establishes existence and finiteness of suitable birational models, from which finite generation of the canonical ring follows. □

**Problem 6.3** (Invariance of plurigenera)**.** Let $\pi : \mathcal{X} \to \Delta$ be a smooth projective family and $X_t = \pi^{-1}(t)$. Must

$$P_m(X_t) = P_m(X_0) \qquad (t \in \Delta,\ m \geq 1)?$$

*Proof.* Siu proved invariance for families of general type in 1998 and for arbitrary smooth projective families in 2002 [117, 118]. Extension of pluricanonical sections, using singular metrics and multiplier ideals, gives one inequality for the dimensions of the spaces of sections; upper semicontinuity gives the reverse inequality. □

*Remark.* Li, Păun, Wang, and Zhou (2026) extend invariance to proper holomorphic submersions with Kähler total space [151, Theorem 1.2]. Liu and Wan's 2026 preprint gives a logarithmic extension: if $\pi : \mathcal{X} \to \Delta$ is a proper holomorphic submersion with connected compact fibres and Kähler total space, and $B = \sum b_i D_i$ is relative simple normal crossings with $b_i \in \mathbb{Q} \cap [0, 1)$, then

$$t \longmapsto h^0(X_t, m(K_{X_t} + B_t)) \text{ is constant}$$
$$\text{for every } m \in \mathbb{Z}_{>0} \text{ with } mB \text{ integral}$$

[152, Theorem 1.1]. Liu and Wan credit ChatGPT as an auxiliary tool and state that they verified and completed the arguments. They remark that, for twisted extension, semipositivity separately on the fibres cannot replace semipositivity on the total space [152, Remark 8.5].

**Problem 6.4.** Let $(M, g)$ be a Kähler manifold with $-b \leq \sec_g \leq -a < 0$. Must

$$H^\infty(M) := \{f \in \mathcal{O}(M) : \sup_M |f| < \infty\} \neq \mathbb{C}?$$

*Proof.* For N. Sibony's formulation in the 1995 problem list [1, Problem 11.3.1], take a compact Riemann surface of genus at least two with a metric of constant curvature $-1$. Every holomorphic function on it is constant by the maximum principle. Thus the displayed curvature condition alone is insufficient. □

**Problem 6.5** (complex structure on $S^6$)**.** Does the standard smooth sphere $S^6$ admit $J \in C^\infty(\mathrm{End}(TS^6))$ such that

$$J^2 = -\operatorname{Id}, \qquad N_J = 0,$$

where $N_J$ is the Nijenhuis tensor?

*Proof.* Claude and L. Alpöge (2026) construct a compact complex threefold $Y$ by completing a family of complex 2-tori over $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ using logarithmic transforms and a Mumford degeneration [141]. P. Engel (2026) gives a self-contained geometric proof that $\pi_1(Y) = 0$ and $H_*(Y;\mathbb{Z}) \cong H_*(S^6;\mathbb{Z})$ [123, Theorem 4.1]. Hence $Y$ is a homotopy 6-sphere; the Kervaire–Milnor calculation $\Theta_6 = 0$ gives a diffeomorphism $f : S^6 \to Y$. Pulling back the complex structure $J_Y$ gives

$$J = (df)^{-1} J_Y df, \qquad J^2 = -\operatorname{Id}, \qquad N_J = 0.$$

□

*Remark.* P. Engel explains that Claude produced the original manuscript under L. Alpöge's direction. He discloses ChatGPT's assistance with exploration, drafting, and computations, while stating that he independently verified the mathematics and accepts responsibility [123, §§1.3–1.4]. J. Viaclovsky also acknowledges ChatGPT assistance in his subsequent constructions [124, 125].

*Remark.* The complex structure is not unique up to biholomorphism. J. Viaclovsky (2026) constructs a two-parameter family containing no member biholomorphic to an original Alpöge–Claude example [124, Theorem 1.1(iv)]. More strongly, he constructs complex threefolds $X_m \cong_{\mathrm{diff}} S^6$ for every integer $m \geq 12$ [125, Theorem 1.1], with

$$h^{0,1}(X_m) = 2m + 1.$$

Since Hodge numbers are biholomorphic invariants, the $X_m$ are pairwise non-biholomorphic.

### 6.2. **Open problems with partial results.**

**Problem 6.6** (Hartshorne conjecture)**.** Is every smooth subvariety $Y^d \subset \mathbb{P}^N$ with $3d > 2N$ a complete intersection? For a rank-two holomorphic vector bundle $E \to \mathbb{P}^N$, must

$$E \simeq \mathcal{O}(a) \oplus \mathcal{O}(b) \quad \text{for some } a, b \in \mathbb{Z}$$

hold if $N \geq 5$, or if $N \geq 4$ and $E$ is not slope-stable with respect to $\mathcal{O}(1)$?

*Partial proof.* Ellia and Franco (2002) proved the complete-intersection property for smooth codimension-two subvarieties of $\mathbb{P}^6$ of degree less than 74 or lying on a quintic, and for subcanonical threefolds in $\mathbb{P}^5$ lying on a quartic [81]. Horrocks and Mumford (1973) constructed an indecomposable rank-two bundle on $\mathbb{P}^4$ [80]. Benoist and Voisin (2025 volume, published 2026) discuss gaps in earlier unstable-splitting arguments [82, §2.3]. Even the weaker assertion that every non-stable rank-two bundle splits for all sufficiently large $N$ remains conjectural in their account. □

**Problem 6.7.** Let $\pi : \mathcal{X} \to \Delta$ be a proper holomorphic submersion, $X_t = \pi^{-1}(t)$, and $S$ an irreducible Hermitian symmetric space of compact type. Does

$$X_t \cong S \quad (0 < |t| < 1) \quad \Longrightarrow \quad X_0 \cong S$$

hold without assuming that $X_0$ is Kähler?

*Partial proof.* Hwang and Mok (1998) proved the assertion when $X_0$ is Kähler [85]. They control varieties of minimal rational tangents and the resulting holomorphic $G$-structures under deformation, thereby identifying the central fibre with $S$. □

**Problem 6.8.** Assume that $K_X$ is not pseudoeffective. Can one construct, by direct complex-analytic methods, a dominant rational map

$$\mathbb{P}^1 \times Y \dashrightarrow X, \qquad \dim Y = n-1,$$

and hence rational curves through general points of $X$? More generally, can the positive, numerically trivial, and negative contributions of $K_X$ be described by canonical fibrations?

*Partial proof.* Boucksom, Demailly, Păun, and Peternell (2013) proved that a smooth projective variety has non-pseudoeffective canonical bundle if and only if it is uniruled [95]. Their cone duality argument uses the numerical uniruledness criterion of Miyaoka and Mori (1986) [96]. Siu (2025) asks for a direct analytic construction and canonical factorization [4].

Ou (2025) extends this uniruledness criterion to compact Kähler manifolds [153, Theorem 1.1]; his proof retains the projective theorem as an input. For the stronger hypothesis that $-K_X$ is ample, Du, Guo, and Xie (2026 preprint) give an analytic construction. Choose a Kähler metric with $\mathrm{Ric}(\omega) \geq \kappa\omega$, $\kappa > 0$, as is possible on every Fano manifold. Then

$$\forall x \in X \quad \exists u_x \in \mathrm{Hol}(\mathbb{P}^1, X) \text{ nonconstant},$$
$$x \in u_x(\mathbb{P}^1), \qquad \int_{\mathbb{P}^1} u_x^*\omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Their proof alternates area-decreasing deformations of holomorphic discs with enlargement of their domains, then applies compactness and removal of singularities [154, Theorem 1.2]. □

*Remark.* Du, Guo, and Xie attribute the mathematical ideas and overall approach to the authors, and disclose AI assistance with exploration and language editing; they reviewed all mathematical statements [154, AI use disclosure].

*Remark.* OpenAI's 2026 orbifold-Iitaka preprint states logarithmic subadditivity for smooth projective reduced-SNC pairs [156, §1], precisely the hypothesis of its abundance paper [155, Assumption 1.1]. Combining these manuscripts gives the claimed implication for a normal compact Kähler log canonical pair $(X, \Delta)$ with effective rational boundary:

$$K_X + \Delta \text{ analytically nef and } \mathbb{Q}\text{-Cartier} \quad \Longrightarrow \quad K_X + \Delta \text{ semiample}$$

[155, Theorem 1.2]. For smooth $X$, $\Delta = 0$, and nef $K_X$, the claimed conclusion gives an Iitaka morphism $f : X \to Z$ with connected fibres and $mK_X \simeq f^*A$ for an ample $A$; a general smooth fibre $F$ satisfies $mK_F \simeq \mathcal{O}_F$. Thus the combined claim addresses the nef canonical-fibration component without a remaining Iitaka assumption, but not the direct analytic construction when $K_X$ is not pseudoeffective. Independent specialist verification of these AI-generated proofs was not located in the literature checked through 25 October 2026.

**Problem 6.9** (Hodge conjecture)**.** For $0 \leq p \leq n$, is

$$H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X) = \mathrm{span}_{\mathbb{Q}}\{[Z] : Z \subset X \text{ algebraic}, \ \mathrm{codim}_X Z = p\}?$$

Here the intersection is taken in $H^{2p}(X, \mathbb{C})$ and $[Z]$ denotes the cycle class.

*Partial proof.* OpenAI (2026) establishes the conjecture for every complex abelian variety $A$ of CM type [142, §1]. Here CM type means that $\mathrm{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ contains a commutative semisimple $\mathbb{Q}$-algebra of dimension $2 \dim A$. For every $p \geq 0$, the cycle-class map is surjective:

$$\mathrm{cl} : \mathrm{CH}^p(A) \otimes_{\mathbb{Z}} \mathbb{Q} \twoheadrightarrow H^{2p}(A, \mathbb{Q}) \cap H^{p,p}(A).$$

P. van Hoften and J. Sempliner (2026) give a variant proof, reducing to codimension two and constructing algebraic representatives using nonzero cup products of theta classes on Picard modular surfaces [143, Theorem I and §1.2]. The general Hodge conjecture remains open. □

*Remark.* The original OpenAI manuscript was produced using an unreleased internal model [142, 145]. P. van Hoften and J. Sempliner disclose extensive ChatGPT Pro assistance with their human-written proof and distinguish their variant argument from a detailed verification of the original manuscript [143, §§1.3–1.4]. J. S. Milne remarks, in an update dated 14 October 2026, that their exposition confirms that the CM case has been proved [144].

*Remark.* Y.-T. Siu remarks that an energy-minimization approach requires a parametrization of a representative of the Hodge class by a compact Kähler manifold. He proposes using multivalued harmonic maps from $\mathbb{P}^{n-p}$ to address this difficulty, and identifies the technical problems caused by multivaluedness [4, §1.5].

### 6.3. **Wide-open problems.**

**Problem 6.10.** For every $N > 2$ and $M > 0$, does there exist a smooth hypersurface $Y \subset \mathbb{P}^N$ such that

$$\operatorname{diam}(Y, g_{\mathrm{FS}}|_Y) > M,$$

where the diameter is computed using the intrinsic Riemannian distance?

*Remark.* J. E. Fornæss and N. Sibony remark that Bogomolov proved the plane-curve case, motivating their question in higher dimension [2, Question 3.12]. Bogomolov's construction is published in 1994 [71].

## 7. Period maps and arithmetic geometry

Period maps encode complex varieties by the Hodge structures on their cohomology. Their images lead to recognition problems, while algebraic varieties defined over number fields lead to questions about rational points and transcendence. Write $\mathcal{A}_g$ for the moduli space of principally polarized complex abelian varieties of dimension $g$, $\mathcal{J}_g$ for its locus of Jacobians of smooth curves, and $\theta(z, \tau)$ for the Riemann theta function. For a number field $K \subset \mathbb{C}$, $X(K)$ denotes the set of $K$-rational points; $\overline{\mathbb{Q}} \subset \mathbb{C}$ denotes the field of algebraic numbers.

### 7.1. **Closed problems with complete solutions.**

**Problem 7.1** (Schottky problem)**.** Characterize the indecomposable principally polarized abelian varieties that are Jacobians of smooth curves.

*Proof.* Shiota (1986) proved the following criterion [102]. An indecomposable principally polarized abelian variety with period matrix $\tau$ is a Jacobian if and only if there exist $U, V, W \in \mathbb{C}^g$, $U \neq 0$, and $c \in \mathbb{C}$ such that, for every $z \in \mathbb{C}^g$,

$$u(x, y, t) = 2\partial_x^2 \log \theta(Ux + Vy + Wt + z, \tau) + c$$

satisfies the Kadomtsev–Petviashvili equation

$$(4u_t - 6uu_x - u_{xxx})_x = 3u_{yy}.$$

These are meromorphic identities. The proof reconstructs a spectral curve from the associated commuting differential operators. □

**Problem 7.2** (Hilbert's seventh problem)**.** For $a \in \overline{\mathbb{Q}} \setminus \{0, 1\}$ and $b \in \overline{\mathbb{Q}} \setminus \mathbb{Q}$, does every value of $a^b$ lie in $\mathbb{C} \setminus \overline{\mathbb{Q}}$?

*Proof.* Gelfond and Schneider independently proved the assertion in 1934 [103, 104]; Schneider's cited journal volume is dated 1935. Their arguments construct auxiliary functions with controlled arithmetic coefficients and high-order zeros, then compare analytic upper bounds with arithmetic lower bounds to contradict algebraicity of $a^b$. □

**Problem 7.3.** Let $f : X \to X$ be an endomorphism of a projective variety over a number field $K \subset \mathbb{C}$. Is

$$h_{\mathrm{top}}(f_{\mathbb{C}}) > 0 \quad \Longrightarrow \quad \overline{X(K)}^{\mathrm{Zar}} = X?$$

*Proof.* Faltings (1983) proved finiteness of $C(K)$ for every smooth projective curve $C/K$ of genus at least two [47]. Set $X = C \times \mathbb{P}^1$ and $f = \mathrm{id}_C \times (z \mapsto z^2)$. Then $h_{\mathrm{top}}(f_{\mathbb{C}}) = \log 2$, whereas

$$\overline{X(K)}^{\mathrm{Zar}} \subset C(K) \times \mathbb{P}^1 \subsetneq X.$$

□

## 7.2. **Open problems with partial results.**

**Problem 7.4.** For every $g \geq 5$, choose a level cover $\mathcal{A}_g^{\ell} \to \mathcal{A}_g$ on which theta constants define projective coordinates, and let $\mathcal{J}_g^{\ell}$ be the inverse image of $\mathcal{J}_g$. Find explicit homogeneous polynomials $F_1, \ldots, F_r$ in these coordinates such that

$$\{F_1 = \cdots = F_r = 0\} \cap \mathcal{A}_g^{\ell} = \overline{\mathcal{J}_g^{\ell}}^{\mathcal{A}_g^{\ell}}.$$

*Partial proof.* Farkas, Grushevsky, and Salvati Manni (2021) construct explicit equations whose zero set has the Jacobian locus as an irreducible component [88]. Their construction applies Schottky–Jung proportionalities to quartic theta identities. Grushevsky's 2004 elimination construction and Grushevsky and Xie's 2026 account explain the distinction between this weak solution and equations excluding all extraneous components [89, 90]. □

**Problem 7.5** (Lang's radius conjecture)**.** Let $C/\overline{\mathbb{Q}}$ be a smooth projective curve with $g(C) > 1$, and let $\varphi : D_r \to C^{\mathrm{an}}$ be a universal covering, where $D_r = \{z \in \mathbb{C} : |z| < r\}$ and $r > 0$. If

$$p := \varphi(0) \in C(\overline{\mathbb{Q}}), \qquad d\varphi_0(\partial/\partial z) \in T_p C(\overline{\mathbb{Q}}) \setminus \{0\},$$

must $r \notin \overline{\mathbb{Q}}$?

*Partial proof.* Cohen and Wolfart (1990) proved the assertion for Shimura curves with a CM base point. Gao, Ullmo, and Yafaev (2026) give another proof of this case [91, §8.1 and Proposition 8.1]: the normalized covering derivative is expressed using CM periods, and transcendence of the resulting period quotient forces the covering radius to be transcendental. □

## 7.3. **Wide-open problems.**

**Problem 7.6.** Let $X \subset \mathbb{P}^N_K$ be a smooth hypersurface over a number field $K \subset \mathbb{C}$, of degree sufficiently large relative to $N$. Does

$$X_{\mathbb{C}} \text{ Kobayashi hyperbolic} \quad \Longrightarrow \quad \#X(K) < \infty?$$

*Remark.* Y.-T. Siu remarks on the Lang–Vojta parallelism between the absence of entire curves and finiteness of rational points. Motivated by jet-differential proofs of hyperbolicity, he proposes a difference-operator analogue of holomorphic jet differentials vanishing on an ample divisor as an approach to arithmetic finiteness [4, §§3.1–3.2].

**Problem 7.7.** For a family of smooth projective varieties with fixed numerical invariants and period map

$$\Phi : S \longrightarrow \Gamma\backslash D,$$

where $D$ is the corresponding period domain and $\Gamma$ the monodromy group, characterize $\Phi(S)$ by explicit differential or algebraic conditions, including any necessary open conditions. For fibres defined over $\overline{\mathbb{Q}}$, determine which algebraic relations among their periods follow from algebraic correspondences.

*Remark.* Y.-T. Siu remarks that polarized Hodge structures provide the higher-dimensional counterpart of period matrices and suggests a Schottky-type characterization by differential or algebraic equations. He points to Griffiths' work on period mappings and connects this discussion with questions about transcendence of periods [4, §§4.5–4.6].

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